



ARBA MINCH UNIVERSITY

COLLEGE OF BUSINESS AND ECONOMICS

DEPARTMENT OF ECONOMICS

INTERNATIONAL ECONOMICS MODULE I

CHAPTER I

1. CHARACTERISTIC FEATURES OF INTERNATIONAL TRADE

CHAPTER OBJECTIVES

The objective of this chapter is to familiarize and acquaint students with the basic characteristic features of international trade. After learning the contents in this chapter, students will be able to:

- *Explain the reasons why countries of the world make trade relations with each other.*
- *Analyze how product price differentials between countries lead to international trade between countries causing one as exporter of the product and the other as importer; identify the factors that cause product price differentials.*
- *Differentiate the static and dynamic gains from international trade; analyze the static (welfare gains) from international trade using graphical models*
- *Explain the difference between international trade and domestic trade in terms of differences in factor mobility, product mobility, economic environment and monetary units.*

1.1 Basis of International Trade

The basic question in international trade theory is to explain why nations trade with each other. Countries trade with each other basically for the same reasons that individual people trade with each other. In today's world of unlimited wants, no nation by itself can produce all the goods and services which its citizens require for their consumption. The following points explain why nations trade with each other.

i. Differences in factor endowments

Nature has distributed the factors of production unequally over the surface of the earth. Countries differ in terms of natural resource endowments, climatic conditions, mineral resources and mines, labor, capital, technological capabilities, entrepreneurial and management skills, and other variables that determine the capacities of countries to produce goods and services. All these differences in production possibilities lead to situations where some countries can produce some goods and services more efficiently than others; and no country can produce all the goods and services in the most efficient manner (at the lowest possible cost of production). For example, Japan can produce automobiles or electronic goods more efficiently than any other country in the world; Malaysia can produce rubber and palm oil more efficiently than other countries can do. Their capacity to produce these goods is in excess of their capacity to consume them. Japan and Malaysia can, therefore, export these goods to other countries at relatively lower prices. Brazil, Ethiopia or Thailand can import these goods at a lower price from Japan and Malaysia and in return they can export coffee (Ethiopia) and rice (Thailand). Because Brazil and Ethiopia can produce coffee at much lower production costs and Thailand can produce rice at much lower cost than Japan and Malaysia.

ii. Division of labor (specialization)

Just as there is division of labor among individuals, there could be division of labor among countries of the world. No individual is able to produce all the goods and services that he/she requires to consume and this applies to countries as well. Just as individuals specialize in certain functions, countries specialize in the production of certain goods and services in which they have production superiorities over the other countries. Thus, a country specializes in the production and export of those goods and services over which they have absolute or comparative advantage; and it imports other goods and services in the production of which it has an absolute or a comparative disadvantage over the other countries of the world.

iii. Gains form exchange of goods and services

The basis of international trade is the gains or profits to be made from the exchange of goods and services. If there are no gains to be made, there would be no such trade between countries.

iv. Price differentials

The immediate cause of international trade is the existence of differences in the prices of goods and services between nations. Foreigners buy our goods because they find them cheaper than anywhere else in the world; and we buy foreign goods because we find them cheaper. Price differences can arise due to either differences in supply conditions or differences in demand conditions or due to differences in both.

a) Differences in supply conditions (or production possibilities)

Such differences can arise due to several factors that determine the cost of producing goods and services. These factors include natural endowments of economic resources, the degree of efficiency with which these factors are employed, the level of technology used in production, labor skills, factor abundance, etc.

b) differences in demand conditions

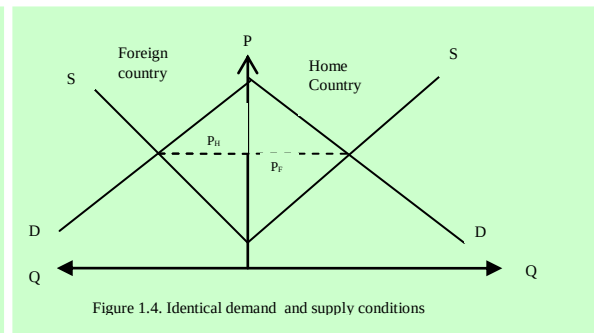
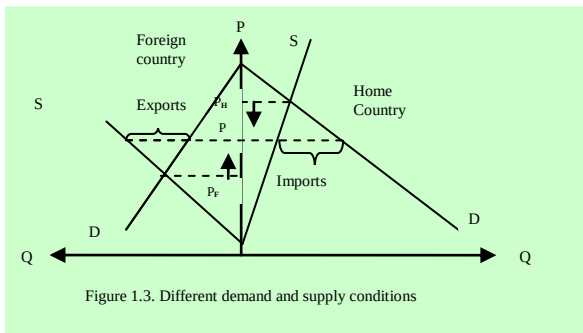
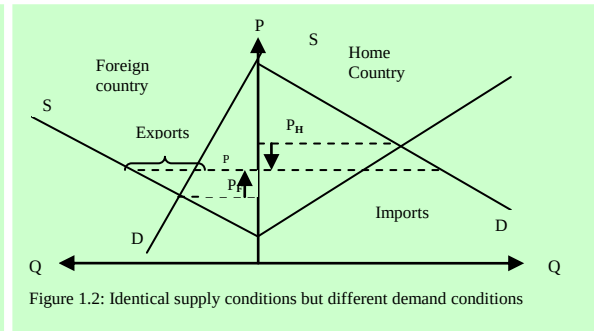
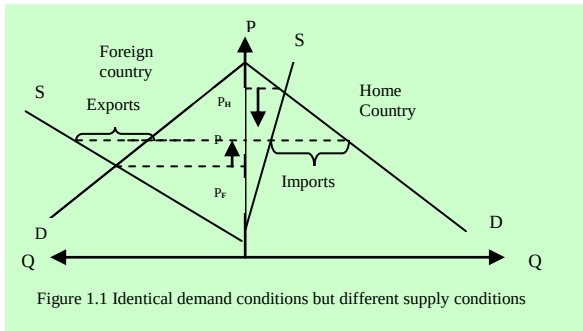
Differences in demand conditions, which are largely a function of income levels and taste patterns, contribute to international price differentials. Countries A and B may be producing the same commodity at the same cost of production, but that does not guarantee the same price in both countries for that commodity. If country A has a higher level of income matched by a stronger taste for that commodity, the commodity will sell for a higher price in country A than in country B. In country B there may be lower level of demand for that product due to lower level of income and/or taste. Hence, both income and tastes, in their capacity to affect demand, would affect international trade, by causing international price differentials.

Let us consider a simple model of the world with two countries producing the same commodity in order to explain the basis of international trade graphically. In all the following three figures the price in the foreign country (P_F) is lower than the price in the home country (P_H) for a given product that both countries produce.

This is due to:

- Differences in supply conditions, where supply curve of the home country is less elastic than the supply curve of the foreign country (figure 1.1).

- differences in demand condition, where the demand curve of the home country is more elastic than the demand curve of the foreign country (figure 1.2).
- differences in both demand and supply conditions, where demand curve of the home country is more elastic than the demand curve of the foreign country, but the supply curve of the home country is less elastic than the supply curve of the foreign country (figure 1.3).



This is in the absence of trade between the two countries. The price differential ($P_H > P_F$) causes international trade between the foreign country and the home country. Higher price in the home country attracts imports of the good into this country from the foreign country where the price of that good is relatively lower. Once the two countries brought into trade relationship with each other, the foreign country would emerge as exporter, and the home country emerge as importer of that product. The trade between the two countries will continue until the price differential is wiped out (i.e. until $P_F = P_H$). When the prices in the home and foreign country are equal there will not be any more incentive for trade to exist between the two countries (i.e. no imports and exports) as

indicated in figure 1.4 above. At this stage the demand and supply conditions in the two countries are identical i.e., the elasticity of the demand functions of the home and the foreign country are the same and the elasticity of the supply functions too.

1.2 The difference between international trade and domestic trade

International trade (external trade or foreign trade) can be defined as the exchange of goods and services between the residents of a given country with those of residents in the rest of the world. Domestic trade (internal trade) is the exchange of goods and services among the residents of the same country. The difference between domestic and foreign trade can be explained as follows:

i. Factor mobility

The distinction made by classical economists between domestic and international trade was grounded on the notion of geographic mobility of factors of production. The assumption made was that within the nation, there is free mobility of factors of production. If internal mobility were perfect, then all factors of production notably labor and capital- would move into areas where they can receive highest rate of return for their services. Under idealized perfect mobility there could not exist inter-regional differences in factor prices. The factors would move away from the regions where their prices are relatively lower to regions where their prices are relatively higher, until the factor price differences between the regions are completely wiped off. In such a case, the price of any factor of production of a given type and quality must be the same throughout the entire country.

In international trade, however, the factor mobility is neither free nor perfect. First of all, there are restrictive immigration laws which prevent free mobility of labor from one country to another. In respect of capital also, there are restrictions on the inflow and outflow of capital and investment across national frontiers. In addition to these legal barriers, there are other social, cultural and political barriers that restrict the mobility of factors from one country to another. Differences in language, climate, social customs and practices, political and educational systems, etc. do create additional barriers to factor mobility between nations. Within the nation, such differences may not exist, or may not appear too formidable to be

overcome by economic incentives. At any rate factor mobility is relatively greater within country than between countries.

ii. Product mobility

Within the nation, the movement of goods and services from one region to another is free. The only internal barriers to free movement of goods and services are the distance and cost of transportation – what may be termed as natural barriers. But in the case of international trade, such a movement is not free, because in addition to the natural barriers, are formidable man-made barriers. For instance, there are import and export duties and quotas, exchange controls, non-tariff (hidden) barriers which put countless obstacles to the free movement of goods and services between countries (we will discuss much about barriers to trade in chapter four). Growing protectionism and spirits of nationalism have been making trade between countries more and more difficult. Agricultural protectionism in the western industrialized countries and the policies of industrialization through import-substitution in the developing countries are some of the examples of how international trade in commodities is being deliberately reduced in today's world.

iii. Economic environment

Within the nation, the economic environment is more or less the same in all the regions of the country. For example, the legal framework or the laws governing consumption, production and exchange of goods and services are the same throughout the country. The government policies with regard to interest rates, taxes, wages or prices are the same within the country. Production techniques, factor proportions, factor prices, infrastructure facilities and production functions or possibilities are nearly the same in the country. Similarly, market structures-the degree of competition or monopoly in production-and consumer taste patterns and preferences are more or less the same throughout the country. All of them would add up to create a certain economic environment or investment climate within the nation. But between nations, they could all differ very significantly. This would make the character of international trade significantly different from that of domestic trade.

iv. Monetary units

The difference between domestic and foreign trade is more obvious in international monetary or currency differences. Within the nation, monetary laws and the financial system and arrangements are the same for all regions in the country. Most significantly, there is a single currency used as a medium of exchange or a measure of value which would make exchange very smooth as far as domestic trade is concerned.

This is not so between countries. We have dollars, euros, yens, pounds, marks, francs, birr etc., and not all of them are freely accepted in discharge of international monetary obligations. An Ethiopian importer must first obtain US dollars before he can think of buying goods from the United States of America. But with the domestic currency you can “import” and “export” goods from one region to another in any quantity you wish. There are no currency complications or convertibility problems involved in carrying out domestic trade. In respect of foreign trade, however, there are currency complications, problems of non-convertibility of currencies, exchange controls and restrictions and many other obstacles. International monetary differences, therefore, introduce complications and complexities in international transactions; and these are absent in domestic trade and exchange.

1.3 Gains from international trade

International trade brings about improvement in production and promotes economic development in the participating countries. It prevents monopolies. It is beneficial to consumers by providing them new and cheap commodities. It also facilitates international payments. The gains from international trade can be broadly classified into static and dynamic gains.

i. Static gains

Static gains arise from optimum use of the country’s factor endowments or human and physical resources, so that the national output is maximized resulting in increase in social welfare.

Let us assume a simple model of international trade with two countries A and B both producing wheat and cotton. The production possibility curves and indifference curves (used to measure utility or welfare) are shown below in figures 1.5 and 1.6.

Figure 1.5 shows that, before the start of foreign trade, country A would be in equilibrium at point E where the price line PP^1 is tangent to both production possibility curve (PPC) AB and indifference curve IC_1 . The slope of the price line shows the price ratio or cost ratio of the two commodities in country A. TT^1 is the terms of trade line whose slope shows the price ratio at which goods can be exchanged between country A and country B. The TT^1 curve is tangent to the PPC AB of country A at point F. Figure 1.5 shows that at F, country A will produce more of cloth (in which it has a comparative

advantage) and less of wheat than at point E. Taking the pattern of demand in country A, we have the indifference curves IC_1 and IC_2 representing the demand for the two commodities.

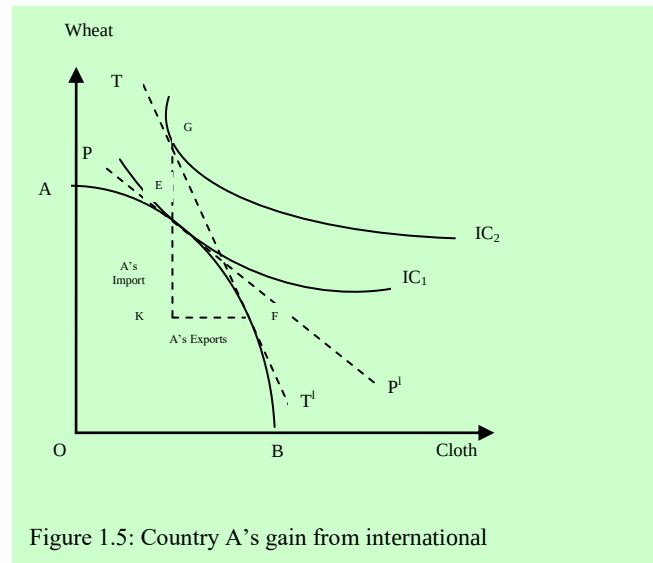


Figure 1.5: Country A's gain from international

The terms of trade line TT_1 is tangent to IC_2 at point G which shows that the quantities of wheat and cloth consumed by country A.

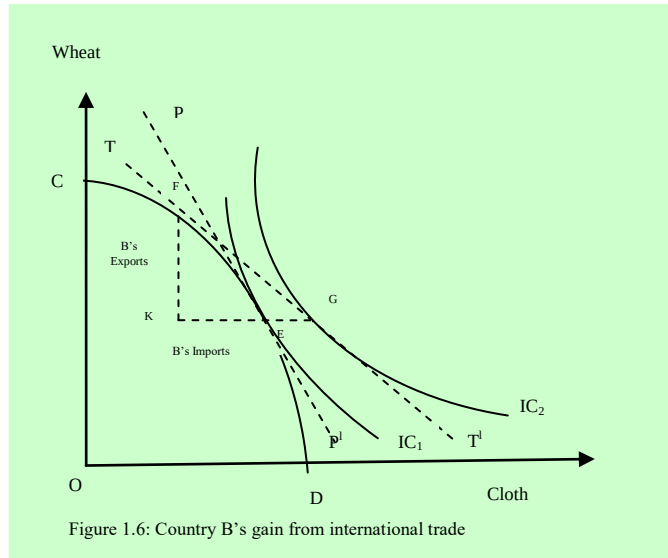
It can be seen that as a result of introduction of international trade, country A has moved from E on IC_1 to G on IC_2 , which represents a higher level of social welfare in terms of larger consumption of the two traded goods. This is called static gain resulting from specialization brought about by the introduction of international trade. It can also be seen that the quantities of the two goods produced are different from the quantities consumed. The quantities produced are shown at F and the quantities consumed at G. The differences are accounted for by exports and imports. Country A will be exporting KF quantity of cloth and importing KG quantities of wheat.

The gains to country B can be similarly explained using figure 1.6 below. Country B will fix its production and consumption at point E before the introduction of international trade.

With the introduction of foreign trade, country B will produce more of wheat in which it has comparative advantage and less of cloth in which it has comparative disadvantage. As a result, it will consume quantities of two goods as shown by point G where the terms of trade line TT_1 is tangent to IC_2 implying a higher welfare level than the situation before foreign trade. Country B will now export KF units of wheat and imports KG units of cloth.

Thus, both trading countries gain from international trade through increased social welfare that can be measured in terms of increased

consumption of the two commodities in both countries.



ii. Dynamic gains

Dynamic gains refer to those benefits, which promote economic growth and economic development of the participating countries. International trade increases national income and facilitates saving and opens out new channels of investment. Increase in saving and investment is found to promote economic growth. Exports earn foreign exchange, which can be utilized in buying capital goods and know-how from abroad that can serve as instruments of economic growth. The larger the national income and output the higher the rate of growth. The higher level of output will enable a country to avoid the vicious circle of poverty and put the country in the take off or self-sustaining growth. Production possibilities and cost of production in different countries differ so widely that foreign trade brings to the participating countries tremendous gains in terms of national output and income.

International trade promotes economic development in the following ways.

- 1) Developing countries can import capital goods in exchange for their exports that are mostly agricultural exports. The capital goods then will increase the productive capacity of these countries and promote the process of industrial development.
- 2) A country can also import technical know-how, technical skills, managerial talent and entrepreneurship through foreign trade and collaboration.
- 3) International trade has brought about a tremendous movement of capital from developed countries to developing countries. Thus, foreign trade facilitates the payment of interest or repatriation of capital. The existence of large volume of foreign trade serves as a guarantee for the payment of interest and the principal for lenders.

CHAPTER SUMMARY

- *In today's world of unlimited wants, no nation by itself can produce all the goods and services which its citizens require for their consumption*
- *Differences in factor endowments, division of labor or specialization in production, static and dynamic gains from international trade, and the existence of price differences among countries are the reasons that explain why nations of the world trade with each other.*
- *Differences in demand conditions, which are largely a function of income levels and taste patterns, as well as difference in supply conditions, which are a function of factor endowments and hence cost of production, contribute to international price differentials.*
- *The difference between domestic and foreign trade can be explained in terms of differences in degree of product and factor motilities in domestic and foreign trade, differences in economic environment under which domestic trade and foreign trade operate, and differences in monetary unit requirements for domestic trade and foreign trade.*

REVIEW QUESTIONS

1. Explain graphically how differences in demand and supply conditions between two hypothetical countries that produce wheat and cloth will lead to trade between the countries.
2. Given that the demand for good X is more elastic in country A than country B while the supply conditions are the same, answer the following questions.
 - a) Compare the before trade price of X in country A and B
 - b) Which country will be importer of X and which country will be exporter?
 - c) What will happen to the price of X in country A and country B during the process of international trade?
 - d) What does the price change in country A result on quantity demand, quantity supply of domestic producers, and supply?
 - e) What does the price change in country B result on quantity demand, quantity supply of domestic producers, and demand?

Note: Answer all of the above questions based on graphical illustration.

3. Give examples of the dynamic gains from international trade.
4. Discuss the static gains from international trade using graphical illustration of how welfare improvements are possible in two countries due to trade.
5. Give examples of barriers to factor mobility in case of domestic trade, factor mobility in case of foreign trade and compare the degree of their effect on factor mobility.
6. What does it mean by an economic environment? How this makes foreign trade different from domestic trade?

CHAPTER II

2. THE CLASSICAL THEORY OF INTERNATIONAL TRADE

CHAPTER OBJECTIVES

The objective of this chapter is to acquaint students with the basic elements of the classical theory of international trade. After learning the contents in this chapter, students will be able to:

- *Explain the basic ideas of Adam Smith's absolute cost advantage theory of international and David Ricardo's Comparative Advantage theory of international trade; compare the similarities and basic differences of the two models of international trade theories..*
- *Understand and analyze how terms of trade between trading countries are determined by demand and supply forces as stated by John Stuart Mill's theory of Reciprocal Demand using the offer curve analysis.*
- *Explain the factors that cause changes in terms of trade between countries and analyze using the offer curve analysis technique.*
- *Compare the Law of reciprocal demand with the theories of absolute and comparative advantages*
- *Critically evaluate the validity of the classical theories of international trade to the real world trade relations between countries.*

2.1. Pre- Classical theory of International Trade (Mercantilism)

The economic philosophy that prevailed during the 17th and 18th centuries was that of the Mercantilism. The main feature of the mercantilist doctrine was that a country could grow rich and prosperous by acquiring more and more precious metals especially gold, and, therefore, all the efforts of the state should be directed to such economic activities that help a

country to acquire more and more precious metals. According to the mercantilist school of economists, if international trade is not properly regulated then people might exchange gold for commodities of daily use or require for a luxurious living. This would lead to the depletion of the stock of precious metals within the nation. Thus, exports were viewed favorably so long as they brought in gold but imports were looked at with apprehension as depriving the country of its true source of riches, i.e., precious metals.

Taxing imports was often justified as a way of creating jobs and income for the national population. Imports were supposed to be bad because they had to be paid for, which might cause the nation to lose spices (gold or silver) to foreigners if it imported a greater value of goods and services than it sold to foreigners. Imports were also to be feared because those same foreign goods might not be available in time of war.

2.2. *The classical theory of international trade*

Adam Smith (1723 – 1790) provided the basic building block for the construction of the classical theory of international trade. He enunciated the theory in terms of what is called Absolute Advantage model. Another well known classical, David Ricardo (1722 – 1823), articulated it and expanded it further into what is called Comparative Advantages model. The models of Smith and Ricardo together constitute what is sometimes referred to as the supply version of the classical theory of trade, because Smith and Ricardo paid almost exclusive attention to considerations of supply or production costs in the determination of terms of trade and the gains from trade. The modern version of the classical theory of trade, however, treats supply and demand with equal weight. John Stuart Mill (1806 -1873), another renowned classical economist, was the first to indicate that demand considerations must be incorporated into Comparative advantage model. But Mill was not very clear or articulate. The vagueness in Mill's principle of Reciprocal Demand was removed in the 19th century, first by F.Y. Edgeworth and later by Alfred Marshall. Both Marshall and Edgeworth are credited with originating and developing the theory of offer curves, which is a geometric technique of demonstrating the theory of reciprocal demand. All these contributions of Smith, Ricardo, Mill, Edgeworth and Marshall, put together, would constitute the modern version of the classical theory of comparative advantage, which is the oldest and the most famous model of international trade.

2.2.1 Adam Smith's theory of absolute advantage

Adam Smith challenged the mercantilists' views on what constituted the wealth of Nations, and what contributed to "nation building" or increasing the wealth and welfare of nations. Smith was the first economist to show that goods, rather than gold (or treasure), were the true measure of the wealth of a nation. He argued that the wealth of a nation would expand most rapidly if the government would abandon mercantilist controls over foreign trade. Smith also exploded the mercantilist myth that in international trade one country gains at the cost of other countries. He showed how all countries would gain from international trade through the international division of labor.

Adam Smith, in his *Wealth of Nations* (1776), argued that a country could certainly gain by trading with other nations. Just as a tailor does not make his own shoes but exchanges a suit for shoes, and hence both the tailor and the shoe maker gain by trading, in the same manner, Smith argued that a country as a whole would gain by having trade relations with other countries. According to Smith, if one country has an absolute advantage over another in one line of production, and the other country has an absolute advantage over the first country in another line of production, then both countries would gain by trading. Let's discuss the Smith's model by taking a simplistic world of two countries A and B both producing Rubber and Textile.

Assumptions

1. There are constant returns to scale in the production of both goods in the two countries (i.e. constant marginal opportunity cost conditions).
2. The production possibilities are such that both countries can produce both the goods if they wish.
3. The countries are endowed with X amounts of factors of production such that:
 - a) With X factors of production country A can produce either 100 units of rubber or 50 units of textile, or any other mix of rubber and textile, that satisfies the opportunity cost ratio of 2:1
 - b) With X factors of production country B can produce either 50 units of rubber or 100 units of textile, or some other combinations of rubber and textile subject to the opportunity cost ratio of 1:2

From the above production possibilities (or supply conditions) it is quite clear that country A has an absolute advantage in the production of rubber, and country B has an absolute advantage in the production of textile. This means there is symmetrical factor distribution between the two countries so that there is scope for specialization in production and also a scope for establishing mutually beneficial trade between the two countries.

A. Autarky (closed economy) situation

The table to the right implies that country A produces and consumes 50 units of rubber and 25 units of textile with the given production technology and input supply. The total production, GDP, is 75 units and this is the maximum consumption level if we assume that saving is zero. Country B produces 25 units of rubber and 50 units of textile with its given

technology and input supplies.

Table 1: Production and consumption levels under autarky situation

Country	Commodities in units		Units of output (GDP)	Opportunity cost ratio (R:T)
	Rubber	Textile		
A	50	25	75	2:1
B	25	50	75	1:2
World	75	75	150	

The total production of country B is 75 units and total consumption will also be 75 units if savings are assumed to be zero. For our simple world of two countries, therefore, world production and consumption will be 150 units.

B. The case of open economy

Opening trade provides the two countries an opportunity to specialize in production. It will lead to production and consumption gains. These effects can be seen in the following table. Country A will specialize in the production and export of rubber and B will specialize in the production and export of textile. Trade will enable the countries to realize production and consumption gains.

After trade both countries become richer by 25 units than their situation without trade and this is due to production gain from international trade. The world GNP has also increased from 150 to 200 units. Country A has specialized in the production of rubber and country B has specialized in the production of textile.

Table 2: Production levels after international trade.

Country	Commodities		Total output or GDP
	in unit		
	Rubber		
	Textile		
A	100	0	100
B	0	100	100
World	100	100	200

What about consumption gains? After trade, have the consumer in the two countries been happier as a result of their countries becoming richer and more specialized in terms of production? This depends on how the gains from production are distributed between the two countries. In other words consumption gains to the two countries depend up on the terms of trade (TOT) i.e. how many units of rubber exchanged for one unit of textile between country A and B.

Case 1. Trade at TOT = 1:1

At this TOT country A agree with country B to exchange 1 unit of rubber for 1 unit of textile. Then depending up on the taste pattern in the two countries and up on how much or how little they want to trade each other's goods, the consumption gains can be determined.

- If the two countries want to consume all that they have produced, it means that their consumers have no taste for the product of the other country then, there will be no trade between countries.
- But if each country wants to consume some mix of both goods; then the countries will trade each other's goods. Country A could export, say; 40 units of rubber for 40 units of textile import from country B (at terms of trade 1:1). The result of this trade can be depicted in table 3 below.

Country A, after trade, has produced 100 units of rubber (see table 2) consumers in country A want to consume 60 of rubber, which means that this country can export 40 units of rubber to country B in exchange for 40 units textile imports.

As a result, their consumption will increase by a total of 25 units than the case without trade (see table 1). Similarly, country B's consumption level will also increase by 25 units than the situation without trade. However, if the terms of trade is equal to the cost ratio of country A, all the gains from trade will be attributed to country B.

Table 3: Consumption shares after international trade at TOT=1:1

Country	Commodities in unit	Total Consumption	Consumption Gains
	Rubber		
	Textile		
A	import 40	100	25
B	import 60	100	25
World	100	200	50

Note: Import of country A is export of country B

Conversely, if the term of trade is equal to the cost ratio of country B, all the gains from trade will be attributed to country A. Any other terms of trade between the cost ratios of the two countries may lead to unequal gains to country A and B. The country whose domestic cost ratio is larger by small amount than the TOT will gain the smaller share and vice versa i.e. if TOT closer to the domestic opportunity cost ratio of country A, country A will gain the smaller and B will gain larger.

Case 2: Trade at TOT = Domestic opportunity cost ratio of country A = 2:1

All the consumption gains from international trade go to country B because the TOT is equal to the domestic opportunity cost ratio of country A. Thus such a TOT is favorable to country B but does not bring any change in welfare level of A.

Table 4: Consumption shares after trade a TOT =2:1

Country	Commodities in unit	Total Consumption	Consumption Gains
	Rubber		
	Textile		
A	50 import 25	75	0
B	import 50	75	50
World	100	100	200

Case 3: Trade at TOT = Domestic opportunity cost ratio of country B = 1:2

All the consumption gains from international trade go to country A because the TOT is equal to the domestic opportunity cost ratio of country B. Thus such a TOT is favorable to country A. It does not bring any change in the consumption level of country B.

Table 5: Consumption shares after trade at TOT =1:2

Country	Commodities in unit	Total Consumption	Consumption Gains
	Rubber		
	Textile		
A	import 75 50	125	50
B	Import 25 50	75	0
World	100 100	200	50

The welfare of country B before and after trade remains constant.

Case 4: Trade at TOT = 2:3

Most of the consumption gains from international trade go to country A because the TOT is closer to the domestic opportunity cost ratio of country B than country A. Thus such a TOT is more favorable to country A than country B.

Table 6: Consumption shares after trade at TOT =2:3

Country	Commodities in unit	Total Consumption	Consumption Gains
	Rubber		
	Textile		
A	import 60 60	120	45
B	import 40 40	80	5
World	100 100	200	50

Case 5: Trade at TOT = 3:2

Most of the consumption gains from international trade go to country B because the TOT is closer to the domestic opportunity cost ratio of country A than country B. Thus such a TOT is more favorable to country B than country A.

Table 7: Consumption shares after trade at TOT =3:2

Country	Commodities		Total	Consumption
	in unit		Consumption	Gains
	Rubber	Textile		
A	40	import 40	80	5
B	import 60	60	120	45
World	100	100	200	50

It is important to note that production gains alone are not sufficient to determine the profitability of international trade from the standpoint of an individual member country. The consumption gains (welfare gains) are also equally important. How the consumption gains are determined is crucial in determining whether the economic well being (standard of living measured by consumption gains) of a member country has gone up as a result of international trade. International TOT, therefore, plays a very important part in determine the welfare gains from trade. International trade would be beneficial and profitable for a country if it results in consumption gains. Production gains alone do not bring profitable trade from the stand point of the country concerned, but it may from world point of view.

2.2.2 David Ricardo's comparative advantage model

Ricardo, in relation to Adam Smith's absolute cost advantage model of international trade, went farther, and argued that even if the countries did not have absolute advantage in any line of production over the others, international trade would be beneficial; leading to gains from trade to all the participating countries. Ricardo's model is termed as comparative advantage model. Ricardo's modal is a further refinement of Smiths' model.

Assumption of Ricardo's Model

- 1) A simple model of the world with two countries each producing two goods, rubber and textile.
- 2) Both countries can produce both goods if they wish (i.e., dependence on each other is not mandatory)

- 3) Constant returns to scale in production of both goods in the two countries (constant marginal opportunity cost condition)
- 4) One country's comparative advantage is greater in one line of production and the other country's comparative disadvantage is smaller in the other line of production.

In simple example the countries are endowed with X amount of factors of production such that :

- a) With x factors of production A can produce, say, 120 units of rubber or 120 units of textile or any mix of rubber and textile conditioned by the opportunity cost ratio of 1:1. The cost of producing a unit of either commodity is the same in this country.
- b) With X factors of production B can produce either 40 units of rubber or 80 units of textile or any mix of rubber and textile conditioned by the opportunity cost ratio of 1:2. Producing rubber costs two times producing textile.

From the above production possibilities (or supply condition) it is quite clear that country A has an absolute advantage over country B in both lines of production and country B has an absolute disadvantage over country A in both lines of production. In terms of relative or comparative advantage, country A has a greater comparative advantage in the production of rubber as compared with the production of textile. B's comparative disadvantage is smaller in the production of textile compared with the first line of production (rubber). Using a given quantity of factor inputs country A can produce 3 units of rubber whereas country B produces only 1 unit of rubber using the same unit of factor input. In case of textile production country A can produce 1.5 times what country B can produce using the same quantity of factor input.

In brief, one country's comparative advantage is greater in one line of production, and the other country's comparative disadvantage is smaller in the other line of production. International trade would bring production and consumption gains, when the two countries enter into trade with each other.

Let's see with the help of numerical models, how international trade will benefit countries in a situation where countries differ in both absolute and comparative advantages. In other words, how countries will benefit from international trade in a situation where one country

has a larger comparative advantage in the production of one good and the other country has a smaller comparative disadvantage in the production of the other good

Table 8: Production possibilities in county A and B

Country	Commodities in		Opportunity cost ratio (R:T)
	Rubber	Textile	
A	120	120	1:1
B	40	80	1:2
World	160	200	

Absolute advantage

- Country A has absolute advantage in the production of both textile and rubber over country B.
- Country B has absolute disadvantage in the production of both goods over country A.

Relative (comparative) advantage

- Country A's comparative advantage over country B is greater in the production of Rubber (3:1) as compared to Textile (1.5:1)
- Country B's comparative disadvantage, in relation to country A, is lower in the production of Textile (1:1.5) as against Rubber (1:3)

In addition country B can produce rubber at a far higher cost of production than textile. For country B the cost of producing 1 unit of rubber equals the cost of producing 2 units of textile. Hence country A would specialize in product of rubber and country B in production of textile. The theory of comparative advantage suggests that a country should specialize in the production and export of those goods in which either its comparative advantage is greater or its comparative disadvantage is smaller. It should import those goods, in the production of which its comparative advantage is smaller or its comparative disadvantage is greater there by a country would be able to maximize its production (GNP) and consumption (economic welfare) gains from trade.

Case 1 Autarky Situation

Table 9 indicates that country a produces and consumes 80 units of rubber and 40 units of textile with the given production technology and input supply. The total production, GDP, is 120 units and this is the maximum consumption level if we assume that saving is zero.

Table 9: Production and consumption under autarky

Country	Commodities		in Total output or GDP
	unit		
	Rubber	Textile	
A	80	40	120
B	20	40	60
World	100	80	180

Country B produces 20 units of rubber and 40 units of textile with its given technology and input supplies. The total production of country B is 60 units and total consumption will also be 60 units if savings are assumed to be zero. For our simple world of two countries, therefore, world production and consumption will be 180 units.

After trade country B becomes richer by 20 units than the autarky situation and the world GNP increases from 180 units to 200 units. This is due to the fact that trade enables country A to specialize in the production of Rubber, for which it has a greater comparative advantage, and country B has specialized in the production of textile for which it has a smaller comparative disadvantage.

Table 10: Production levels after internal trade

Country	Commodities		Total and Consumption	GDP	GNP gains
	unit				
	Rubber	Textile			
A	120	0	120		0
B	0	80	80		20
World	120	80	200		20

As in the case of Smith's model, the distribution of production gains from international trade among trading countries for consumption depends on the terms of trade. The following tables illustrate the distribution of consumption gains at different terms of trade.

The gains are equally distributed to country A and B at $TOT = 3:4$. This TOT is exactly half way between internal cost ratios of country A (1:1) and country B (1:2). From table 10 and 11 we observe that, although the production gains are obtained entirely by country B, the consumption gains are distributed equally between country A and B. This is due to the equitable terms of trade.

Table 11: Consumption levels after IT at $TOT = 3:4$

Countr y	Commodities in unit		Total Consumpt ion	Consumptio n Gains
	Rubber			
	Textile			
A	90	import 40	130	10
B	import 30	40	70	10
World	120	80	200	20

All the gains in consumption are appropriated by country B due to the reason that TOT is equal to the internal opportunity cost ratio of country A. In this situation, country A will not gain anything from international trade, because the cost of importing the good is the same as the cost of producing that good domestically.

Table 12: Consumption levels after IT at $TOT = 1:1 = OCR$ of A

Country	Commodities in unit		Total Consumption	Consumption Gains
	Rubber			
	Textile			
A	80	import 40	120	0
B	import 40	40	80	20
World	120	80	200	20

All the gains in consumption are appropriated by country A due to the reason that TOT is equal to the internal opportunity cost ration of country B.

Table 13: Consumption levels after IT at TOT = 1:2 = OCR of B

Country	Commodities in		Total Consumption	Consumption Gains
	unit			
	Rubber	Textile		
A	90	import 60	140	20
B	import 30	20	60	0
World	120	80	200	20

2.2.3 Law of Reciprocal Demand Offer curve Analysis

The Principle of Reciprocal Demand was developed by J.S. Mill in 1848 when he wrote his book: Principles of Political Economy. Later offer curve technique was developed by Edgeworth and Marshall. The offer curve tries to show how the terms of trade are determined by the interaction of demand and supply. Their merit lies in the fact that they resolve the problem of determining the exact terms of trade that emerge in trade equilibrium.

Assumptions

- Simple model of the world with two countries A and B,
- Country A specializes in the production of textile and B in the production of rubber.

The TOT, then according to the law of reciprocal demand, are determined by A's demand for B's product and B's demand for A's product. In other words, the TOT are determined by the intensity of domestic demand for foreign goods (offer curve of the country A) and of the foreign demand for domestic good (offer curve of the other country B). The equilibrium TOT is determined at the point where the offer curves of the two countries intersect.

Table 14 indicates that textile is country A's exportable product, and rubber is its importable product. Country A has completely specialized in the production of textile, because it has an absolute or comparative advantage in this line of production. If consumers in country A want to consume rubber, it can be fulfilled only by importing it from country B, which has achieved complete specialization in rubber production.

Table 14: Schedule of Trade Propensity for country A

Textile (Exportable)	Rubber (Importable)	Potential Terms of Trade ratios
25	5	5:1
40	10	4:1
60	20	3:1
80	40	2:1
100	100	1:1
90	120	0.75:1

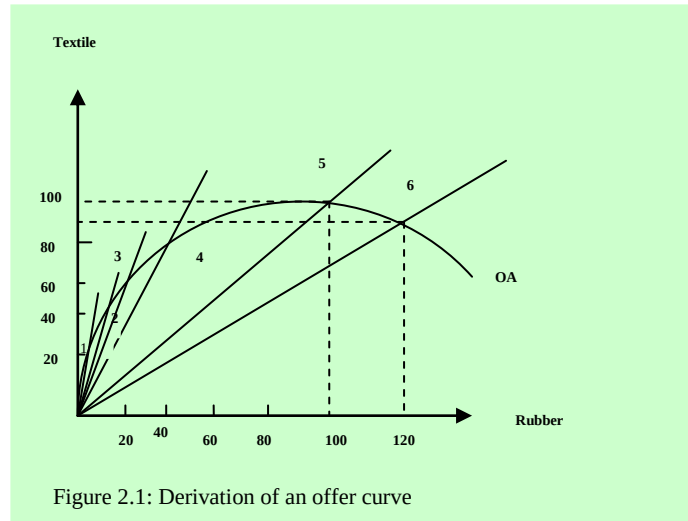
Initially when country A does not have any rubber to consume at all, it is willing to export 25 units of textile in exchange for 5 units of rubber implying trade takes place at 5:1 terms of trade.

This is the highest TOT in table 5 and as trade continues between A and B, the TOT declines. The reason is, initially the marginal utility of rubber for country A is so high that it is willing to offer 5 units of textile (export) in exchange for 1 unit of rubber (import). But as trade continues and country A consumes more and more of the imported product (rubber), the marginal utility from additional consumption of rubber goes down. This is explained by the decrease in the TOT ratios in column 3 of table 10 from 5:1 to 1:1.

This follows from the law of diminishing marginal utility where the consumer is willing to pay higher price per unit of utility initially, but only lower and lower price for subsequent unit that he/she buys. In the same way the price that a country is willing to pay per unit of imports decrease as the imports increase in quantity. The TOT in column 3 of table 14 are potential terms of trade, which only indicate country A's propensity to trade more and more at different possible terms of trade. That is, as TOT decreases what country A want to export and import will also increase as indicated in columns 1 and 2 of table 14. We can plot the information in table 10 and draw the offer curve of country A as shown below.

The continuous line which is drawn from the point of origin connecting all the five points, 1,2,3,4, and 5 is the offer curve of country A. It is designated as OA. It has a positive slope and it is non-linear. Its positive slope derives from country A's desire to trade more and more, though at different terms of trade. Up to point 5 its slope continues to be positive. At point 5, country A is willing to offer 100 units of textiles in exchange for 100 units of rubber imports. Beyond point 5,

however, the offer curve has a negative slope. This indicates that at point 5 where country A exports 100 units, its desire or the capacity to export more textiles is exhausted.



At point 6 in the diagram, country A is willing to accept or import 120 units of rubber at a price of only 90 units of textile exports. This will not be acceptable to country B; because, for country B it is better to sell 100 units of rubber and receive 100 units of textiles at point 5 than trading at point 6. This means country A would be willing to trade with country B even beyond point 5, but this will not be acceptable by country B. Therefore, trade virtually cannot take place beyond point 5. Thus, we can have the following properties of offer curve of a country.

i. The relevant portion of the offer curve is positively sloped

Only the positively sloping portion of the offer curve is relevant for trade equilibrium. The negative portion of the other offer curve is an irrelevant range.

ii. The offer curve is not a straight line, it is a non-linear curve.

This property follows from the law of diminishing marginal utility for the purchased goods. For a given unit of imported commodity, country A is willing to offer less and less price as it

goes on importing more and more quantity. The rays which are drawn from the point of origin across points 1, 2, 3, 4, 5 and 6 represent potentially acceptable terms of trade from the standpoint of country A. They all have different slopes suggesting that the price of rubber, payable in terms of the number of units of textiles per unit of rubber imported, goes on decreasing as country A imports larger and larger quantities of imports.

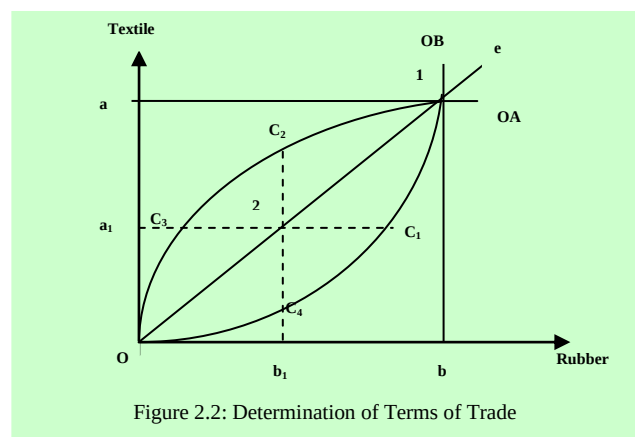
iii. The offer curve is a composite curve (both demand and supply)

Each point on the offer curve represents the demand of the country for imports and supply of the country (exports). The offer curve of a country has an affinity with the potential terms of trade. The question then is where the actual terms of trade are? This cannot be known until the other country's offer curve is known. It is unnecessary to repeat the whole explanation, because the same principles apply to country B's offer curve as those of country A's offer curve.

The only point to emphasize for contrast would be that country A's importable product would be country B's exportable. Once we know the offer curve of country A and country B, we can see where the actual terms of trade are. Figure 2.2 demonstrates where the terms of trade are actually settled between the two countries.

OA and OB are the two offer curves of country A and country B. Both intersect at point 1, country A is willing to offer (export) Oa quantity of textiles in exchange for Ob quantity of rubber imports. This perfectly matches with country B's trading propensity as well, because at point 1, country B is willing to offer (export) Ob quantity of rubber in exchange

for Oa quantity of textile imports. In other words, A's demand for B's product exactly coincides with B's demand for A's product at point 1.



The straight line starting from the point of origin and going through point 1 is the equilibrium terms of trade line, Oe . The slope of this line represents the international terms of trade (i.e., the price of rubber in terms of textile).

The two countries can trade as much or as little as they wish along the terms of trade line Oe , but beyond point 1 on the Oe line they will have no incentive to trade further. The equilibrium size of trade is represented by the triangle $Oa1$ or $Ob1$ i.e., when Oa exports of country A (or $b1$ imports of country B) are matched by $a1$ imports of country a (or Ob exports of country B). Beyond point 1 there will be no incentive for trade between the two countries. By the same token, an incentive to trade will not stop until the two countries reach point 1 on the Oe line. Let us see why this is so. Take for instance a point such as point 2 on the terms of trade line Oe . At point 2 there is:

- **Excess offer of C_3C_1 amount of rubber by country B to country A, and**
- **Excess offer of C_2C_4 amount of textiles by country A to country B.**

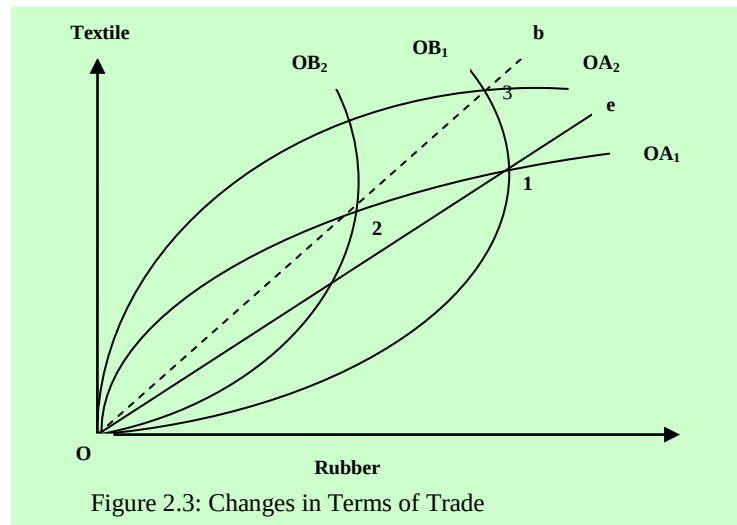
Both the countries accept this excess offers and their trade volumes increase. This process of trade expansion will go on until the excess offers by the two countries are eliminated. Such a point is reached when both countries arrive at point 1, where country A's offer is exactly equal to what country B is asking for, and country B's offer is also exactly equal to what country A is willing to accept. We say that at point 1 the reciprocal demands of the two countries are in exact balance and trade equilibrium point is reached. This has taken place at a point where the two offer curves intersect. The size of international trade corresponding to this equilibrium position is Oa of textile exports plus $b1$ of rubber imports for country A that equals $a1$ textile imports plus Ob rubber exports of country B at Oe terms of trade.

Changes in Terms of Trade

The offer curve of a country is both a demand curve and a supply curve. It is a demand curve to the extent that it represents domestic demand for foreign goods. It is also a supply curve insofar as it represents quantities of exportable goods offered in exchange for imported good. An offer curve of a country, therefore, is a composite curve, representing demand for importable goods and supply of exportable goods. This applies to offer curves of both (or all) the countries. When the offer curves of two countries intersect at a point such as 1 in figure 2.2 we have equilibrium terms of trade in the

static sense. But the conditions underlying demand and supply in the home and the foreign country do not remain static forever. Many dynamic changes are possible. For example, the domestic demand for foreign goods may change due to changes in tastes, incomes, foreign prices of the goods etc. Similarly, the supplies of exportable goods may change on account of changes in domestic cost conditions or demand conditions. In the same way there could be changes affecting foreign country's offer curve as well.

Figure 2.3 demonstrates the effects of such dynamic changes on the equilibrium terms of trade and size of international trade. OA and OB are the offer curves of countries A & B respectively. Point 1 determines equilibrium size of international trade in the static sense; Oe line represents static terms of trade. A leftward shift in country B's offer curve from OB₁ to OB₂ would shift the terms of trade line from Oe to Ob.



This would shift the trade equilibrium position from point 1 to point 2 as shown in figure 2.3. Such a shift causes terms of trade to improve for country B and worsen for country A. Country A is now able to import a smaller quantity of rubber in exchange for a given quantity of textile exports. This is suggested by the fact that the new terms of trade line, Ob, is steeper than the original terms of trade line Oe. As the terms of trade line becomes steeper, and keeps moving closer to the vertical axis (where we measure country A's export product) the terms of trade would go on worsening for country A and improving for country B.

A shift from point 1 to point 2 causing terms of trade deterioration for country A has, in this case come about due to adverse changes in country B for country A's export product. This may have been the as a result of several factors like:

- A drop in the taste in country B for the product of country A.
- Country B might have diverted its trade direction away from country A to some other country in the world. This will reduce country B's demand for country A's product.

- Country B might have launched a program of import substitution as a result of which it has started' producing country A's product domestically within country B itself.

It is also possible for terms of trade to worsen for country A even though there are no adverse changes in country B towards the product of country A. For instance, starting from point 1 we could end up at point 3 in figure 2.3. This will also shift the terms of trade line from Oe to Ob . Note, however, that this time the worsening of terms of trade is a result of a leftward shift in country A's offer curve from OA_1 to OA_2 . Such a shift is due to changes that have taken place within country A itself. These internal changes could be:

- Increase in export production in country A leading to fall in the price of export product.
- A technological revolution in the export production has considerably cut costs of producing textiles in country A. This would mean that country A has become a more efficient producer and exporter of its product to country B.

In this way it is possible to visualize number of changes in supply condition in country A that will have the effect of shifting its offer curve from OA_1 to OA_2 . Compare now the effects on terms of trade on country A as a result of going from point 1 to point 2 and going from point 1 to point 3. In either case, of course, there has been the same degree of deterioration in terms of trade affecting country A adversely. But there are differences too. These differences are:

- Movement from point 1 to point 2 has resulted in contraction of the volume of world trade; but in respect of the movement from point 1 to point 3 there has been an expansion in trade volume.
- Movement from point 1 to point 2 is the result of an adverse demand shift in the foreign country (country B) causing terms of trade decline for country A. Country A may suffer trading losses on account of such terms of trade deterioration. In the case of a movement from point 1 to point 3, however, there is decline in terms of trade for country A; but this term of trade decline may not amount to a trading loss as such. Because, this movement has resulted from favorable supply shifts in the home country (country A) itself, such as for example technological progress or efficiency

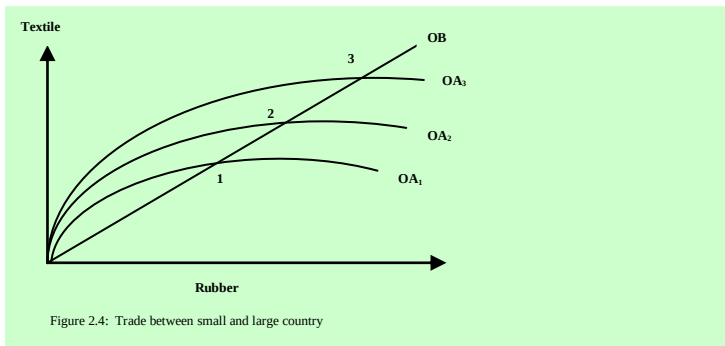
in export production leading to reduced input costs. Country B, in this case benefits from such technological progress in country A, because country A can now sell its product at a lower price to country B. Country A can increase country B's welfare without suffering any loss of welfare itself. In this way the benefits of progress in one country (country A in this case) can be transmitted to other countries (country B) in the form of reduced export prices.

The classical economists had these advantages of trade in their mind; they thought that international trade would act as a "transmission belt" in which the benefits of growth, productivity and efficiency in one country will be passed on to other countries through the mechanism of trade.

The reality of the world today may appear to be quite puzzling. Advanced industrial countries have experienced remarkable progress in technology and productivity in the last several years, but this does not seem to have resulted in reduced prices of their products to the developing countries. Examples are not lacking. According to a United Nations' report, 25 tons of rubber exports could earn enough foreign exchange for a less developed country to buy 6 tractors from an advanced industrial country in 1960; but in 1975, the same 25 tons of rubber can buy only 2 tractors.

Technological breakthrough in rubber production has reflected in rubber price reduction by rubber exporting countries (i.e. the LDCs), but the industrial countries have not reduced their tractor prices in spite of a much more spectacular technological progress in all fields of industrial production. This is quite contrary to what the classical economists had hoped about benevolent trade between countries. Where have the productivity gains, resulting from technological progress in the industrial countries, gone then? They have gone in the form of higher wages, profits and standards of living in those countries themselves. The benefits of technological progress in the LDCs have gone to the industrial countries in the form of reduced prices of primary goods, which the LDCs export to the industrial countries. This would only suggest that trade is benefiting the rich countries and not the poor countries. Productivity gains are going from the poor to the rich countries, while rich countries themselves are retaining their productivity gains by improving their wage and profit levels. LDCs are quite bitter about such kind of trade relationship existing in the world today.

Let us now take another situation where the offer curve of country B, rich country, is a straight line and the offer curve of country A, poor country, not as shown in the following figure 2.4.



To whatever direction the offer curve of country A shifts, the terms of trade remains constant and equal to the offer curve of country B. From this it should be realized that a small country could exercise no influence in changing the course of international terms of trade. A small country, like country A in this case, is essentially price taker than a price-maker.

CHAPTER SUMMARY

- *Mercantilists measure the welfare of a society in terms of the accumulation of precious metals like gold and silver. Thus, they were against free trade in general and favor exports so long as they brought in gold but imports were looked with apprehension as depriving the country of its precious metals.*
- *According to Smith's theory of absolute cost advantage, the wealth of a nation is measured in terms of real goods and services.*
- *According to the theory of absolute cost advantage, a country has to specialize in the production and export of the good for which it has an absolute cost advantage over the other country and import the good for which it has an absolute cost disadvantage over the other country. This will benefit both countries in terms of production and consumption gains.*
- *Ricardo's theory of comparative advantage states that a country has to specialize in the production and export of the good for which it has either a larger comparative advantage or smaller comparative disadvantage over the other country. It has to import the good for which it has either a smaller comparative advantage or a larger comparative disadvantage over the other country.*

comparative disadvantage over the other country. Such a trade relation will benefit both trading countries.

- The distribution of welfare (consumption) gain from international trade among the trading countries depends on the international terms of trade. If the terms of trade is closer to the domestic opportunity cost ratio of one country, most of the welfare gains will be attributed to the other country and vice versa.
- Both Smith's and Ricardo's theories of international trade are termed as supply version of the classical theory of international trade. Both theories paid exclusive attention to production costs in the determination TOT and gains from international trade.
- The law of reciprocal demand and the offer curve analysis resolve the problem of determining the exact terms of trade that emerge in trade equilibrium. According to this law terms of trade are determined by the forces of demand and supply.
- The offer curve is a composite curve in a sense that it represents both demand and supply. Each point on the offer curve represents the demand for imports and the supply of exports of a country. All factors that affect demand and supply of a product do have the same effect on the offer curve of a country and cause it to shift. The shift of the offer curves will cause a change in terms of trade.

REVIEW QUESTIONS

1. Write the statements of the absolute cost advantage and the comparative advantage theories of international trade; compare their differences and similarities.
2. Assuming a simple model of two countries A and B where A is exporter of coffee and B is exporter of steel based on their absolute and comparative advantages answer the following questions.
 - a) If the offer curve of country A is constant but the offer curve of country B shifts right ward due to increase in income of people in country B and positive taste and preference to the product of country B, Coffee, what are the effects of such change on:
 - Equilibrium terms of trade (which product will be cheaper than the basic situation; show it graphically)
 - Volume of world trade
 - Country A and B
 - b) What factors cause the offer curve of country A to shift up showing increase in supply of coffee and down rightwards showing a decrease in supply of coffee at a

given terms of trade.

3. Write the properties of the offer curve of a country.
4. What does the law of Reciprocal demand states about and how does it differ from the theories of Smith and Ricardo?
5. Suppose that the following table shows the case of an open economy of comparative advantage model of two countries Japan and China. Both countries can produce cloth and steel at different internal domestic cost ratios. The production and consumption levels in the case of autarky for the two countries are given in the following tables.

Table 1: Production schedule

Countries	Commodities produced in units		
	Cloth	or	Steel
Japan	240	or	240
China	80	or	160
World	320	or	400

Table 2: production and consumption under autarky

Countries	Commodities produced and consumed in Unit		Total production and consumption
	Cloth	Steel	Cloth + Steel
Japan	160	80	240
China	40	80	120
World	200	160	360

Based on the above information answer the questions a to d.

- a) Compare the two countries in terms of both absolute and relative advantages
- b) Calculate the internal opportunity cost ratios for each country and determine in which production line each country should specialize according to the comparative advantage theory.
- c) What is level of the production gain due to specialization and trade
- d) What will be the consumption gain between china and Japan if the international TOT = 1:1. Based on the result give general statement on how TOT determines the distribution of welfare gains from international trade among trading countries.

CHAPTER THREE**3. THE MODERN THEORY OF INTERNATIONAL TRADE****CHAPTER OBJECTIVES**

The objective of this chapter is to acquaint students with the two main propositions of the modern theory of international trade. After learning the contents in this chapter, students will be able to:

- *Analyze the factor endowment theory (Heckscher-Ohlin theorem) and its predictions about the structure of international trade based on the price and physical criteria of defining factor abundance..*
- *Compare the validity of the H-O theorem under price and physical criteria.*
- *Critically evaluate the validity of the H-O theorem based on the factor intensity reversal argument, the demand reversal argument and the Leontief's paradox.*
- *Analyze the basic principles and ideas of factor price equalization theorem base on the Edgeworth-Box graphical analysis.*
- *Critically evaluate the validity of the factor price equalization theorem by comparing its assumption with the empirical world.*

The two main propositions of the modern theory of international trade are the Factor-Endowment theory (Heckscher-Ohlin theorem, hereafter named H-O theorem in this module) and the Factor-Price equalization theorem.

The Factor-Endowment theory (H-O Theorem) states that a country has a comparative advantage in the production and exports of that commodity which uses more intensively the country's relatively abundant factor of production.

The Factor-Price Equalization Theorem states that the effect of trade is to equalize factor prices between countries, thus serving as substitute for international factor mobility.

3.1 The H-O Theorem (Factor-Endowment theory) and its Assumptions

Recent contributions to the pure theory of international trade have relied heavily on the factor proportions analysis developed by the two Swedish economists, Eli Heckscher (1919) and Bertil Ohlin (1933). According to their theory, the immediate cause of international trade is, the differences in the relative prices of commodities between the countries, and

these differences in the commodity prices arise on account of the differences in the factor supplies (endowments) in the two countries.

The H-O model is based upon the following assumptions:

- There are only two factors of production-labor and capital.
- There are only two countries and they are different in factor abundance, e.g. one country is capital abundant but labor scarce and the other country is labor abundant but capital scarce. In other words, the two countries differ in factor endowments.
- There are only two commodities. Both goods involve the use of both factors. The production functions are such that the relative factor intensities are the same for each good in the two countries. In other words, regardless of what the factor proportions or factor prices are in the two countries, one commodity is always capital intensive in both countries and the other commodity is labor intensive in both countries.

On the basis of these assumptions, the H-O theorem predicted that the capital surplus country specializes in the production and exports of capital intensive goods, and the labor surplus country specializes in the production and exports of labor intensive goods. We will now proceed to demonstrate this well-known structure of trade prediction of the H-O model. In order to demonstrate this prediction we need to define the term factor abundance.

3.1.1 Factor Abundance

There are two alternative definitions that have been given for the term 'factor abundance' based on the criterion used to define the term. There are two criteria for defining the term factor abundance. These are price criterion and physical criterion.

i. The Price Criterion

According to this "price criterion" a country in which capital is relatively cheap and labor is relatively more expensive, is regarded as the capital abundant country, regardless of the physical quantities of capital and labor available in this country compared with the other country; in the same way, a labor abundant country would be defined as one where labor is relatively cheaper and capital is more expensive.

For the fact that price of a factor is the result of demand and supply forces in the factor market, this criterion takes into account the supply and demand conditions for the two factors of production in the two countries.

ii. The Physical Criterion

Factor abundance can be defined in physical terms. According to the "physical criterion", a country is relatively capital abundant if and only if it is endowed with a higher proportion of capital to labor than the other country. In other words, if the capital to labor ratio of country A is larger than the capital to labor ratio of country B, country A is a capital surplus country and country B is a labor surplus country. This criterion takes into account only the supply (physical quantities) of factors as a base for defining factor abundance. It does not take factor demand into account.

These two alternative definitions are not equivalent. The Heckscher-Ohlin prediction with regard to the structure of trade would follow only if we use the price criterion but it does not necessarily hold well if we use the physical criterion to define factor abundance. Ohlin himself defined relative factor abundance using the price criterion. He thought that if capital is relatively cheap in one country, that country must be abundant in capital supply; and if labor is relatively cheap in the other country, it must be a reflection of the labor abundance in that country.

It now remains for us to show that one country, say country A, is capital abundant and it exports capital-intensive good, and the other country, say country B, is labor abundant and, therefore, it will export labor intensive good. We shall examine these Heckscher- Ohlin proposition about the structure of trade under each definition of factor abundance separately.

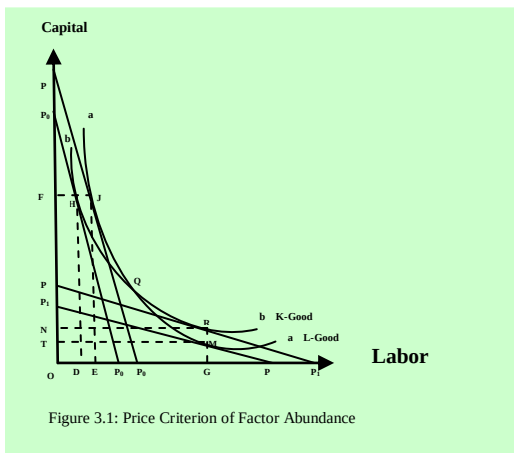
3.1.1.1 Price Criterion of Factor Abundance and the Structure of trade

Starting from the definition of factor abundance in terms of factor prices, it is easy to establish the H-O theorem. It is easily demonstrated in figure 3.1. The two factors-capital (K) and labour (L) are measured along the vertical and horizontal axis, respectively. The set of factor price ratios (isocost curves) in country A, a capital surplus country, are shown by the parallel isocost lines P_0P_0 and $P_0'P_0'$. The slope of the isocost curve shows the factor price ratios in country A ($\text{Slope of Iso cost curve} = \frac{\Delta K}{\Delta L} = \frac{P_L}{P_K}$). The relative steepness of these

lines reflects the fact that capital is cheaper and labor is dearer in country A. Similarly, the

lines P_1P_1 and $P_1'P_1'$ reflect the factor price ratios (isocost curves) in country B. The relative flatness of these lines shows that labor is cheap and capital is expensive in country B, a labor surplus country.

Then we have the two isoquants labeled aa and bb, and the two isoquants cut each other only once, i.e. at point Q. This indicates that there is no reversal of factor intensity, meaning that one commodity is capital intensive in both the countries (K good represented by the isoquant aa) and other commodity (Good represented by the isoquant bb) is labor intensive in both countries. This is in conformity with the H-O assumption that the production functions are identical for each good in the two countries. We can now see how the capital surplus country would export capital-intensive good, and labor surplus country would export labor intensive good.



i. Cost of Production in Country A

The cost of producing one unit of K good is made up of HD amount of capital plus HF amount of labor, because at point H there is a tangency between the isocost curve P_0P_0 and the isoquant for K good. The cost of producing one unit of L good consists of $OF=JE$ amount of capital (which is equal to the capital unit required to produce the K-good) but more labor viz. OE amount of labor (as against OD amount needed to produce one unit of K good). In other words, in country A, in order to produce one unit of L good, you need to use the same amount of capital as in K good (viz. $HD=JE = OF$) but more labor (OE as against OD). In other words, the isoquant curve of K-good is tangent at the lower isocost curve whereas the isoquant curve of L-good is tangent to the upper isocost curve of country A. This implies producing K-good is cheaper than producing L-good in country A. Hence the capital surplus country (country A) would specialize in the production and exports of capital-

intensive good (K good). K-good is capital intensive and L-good is labor intensive to both country A and B due to a higher capital to labor ratio in K-good production than the L-good production as shown in figure 3.1.

$$\frac{OF}{OD} (K - \text{good}) > \frac{OF}{OE} (L - \text{good}) \text{ for Country A and } .$$

$$\frac{ON}{OG} (K - \text{good}) > \frac{OT}{OG} (L - \text{good}) \text{ for country B}$$

ii. Cost of Production in Country B

The cost of producing one unit of L good is made up of OT = MG amount of capital plus OG amount of labor, but the cost of producing one unit of K good consists of the same amount of labor but more amount of capital i.e. ON = RG (as against OT=MG needed to produce one unit of K good). This means that country B can produce L good at a relatively lower cost of production per unit. Therefore, country B (a labor surplus country) would specialize in the production and exports of L good (a labor intensive good).

To sum up:

- a) Factor price ratios in country A and B are different, which reflects that country A is capital-abundant and country B is labor-abundant,
- b) One commodity is capital intensive in both countries (viz. K good) and the other commodity is labor intensive in both countries (L good).

Thus, starting from the definition of factor abundance in terms of factor prices, (or price criterion) it is easy to establish the H-O theorem. Incidentally, we might also say that reverse of the theorem also holds well, i.e. if a country exports capital intensive good, then capital must be its cheaper factor of production. Likewise, if a country exports labor-intensive good, then labor must be a cheaper factor of production in that country.

One could argue, however, that stating the theorem in terms of factor prices, is not very interesting, because factor prices are themselves the result of a complicated interplay of economic forces- particularly of demand for and supply of the factors of production. From the mere knowledge of factor endowments alone it is not possible to say anything precisely about factor prices. Therefore, to state the H-O theorem in terms' of factor prices (using the price-criterion) does not give, perhaps, the most interesting explanation of the theorem. A more natural definition of factor abundance, it appears would run in terms of physical amounts. We shall now turn to the physical criterion and see what the results would turn out to be:

3.1.1.2 Physical criterion of factor abundance and the structure of trade

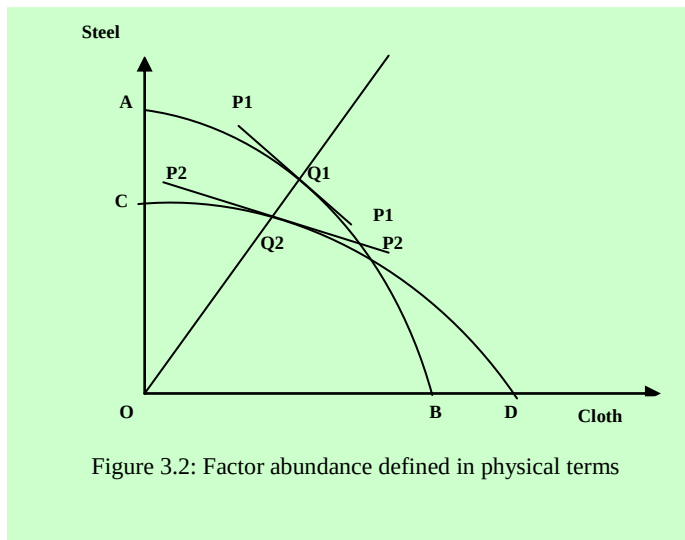
According to this criterion country A is capital abundant and B is labor abundant if the capital to labor ratio of country A is greater than the capital to labor ratio of country B.

Symbolically: $\frac{K_A}{L_A} > \frac{K_B}{L_B}$

Where K_A and L_A are the total amounts of capital and labor, respectively, in country A, and K_B and L_B are the amounts of, capital and labor, respectively, in country B.

We will now show that country A, a capital abundant country by the physical criterion of abundance, has a bias in favor of producing the capital-intensive good; and that country B, a labor abundant country will have a production bias in favor of labor-intensive good production.

Figure 3.2 reflects the nature of these biases in the two countries in respect of the two goods. The production possibility curve of country A is AB and that of country B is CD. We assume that steel is the capital intensive good and Cloth is the labor intensive good. Suppose, the two countries produce the goods in the same proportion-along the ray OR-then country A would produce at Q 1 and country B at Q2 on their respective production possibility curves. Note that the slope of country A's production-possibility curve at Q 1 is steeper than the corresponding slope of country B at Q2.



Similarly, the commodity price line P_1P_2 is steeper than the line P_2P_1 . All this implies that steel production is cheaper in country A and cloth production is cheaper in country B, if the two countries are producing at Q_1 and Q_2 respectively. Country A would, therefore, tend to expand production of steel and country B would do so for cloth. This means that country A has a bias in producing more steel than cloth and B has a bias in producing more cloth than steel.

Does it follow from this that country A would export steel and country B would export cloth? The answer depends very much upon the demand (consumption) factors in each country. This gives rise to two possibilities:

- If the consumption bias and the production bias are towards the same direction, then country A would import rather than export steel and country B would import rather than export cloth. The H-O prediction would then be invalid,
- If the consumption and production biases are in the opposite direction, then the H-O prediction will be valid, viz. country A would export steel and country B would export cloth. Let us illustrate these two cases.

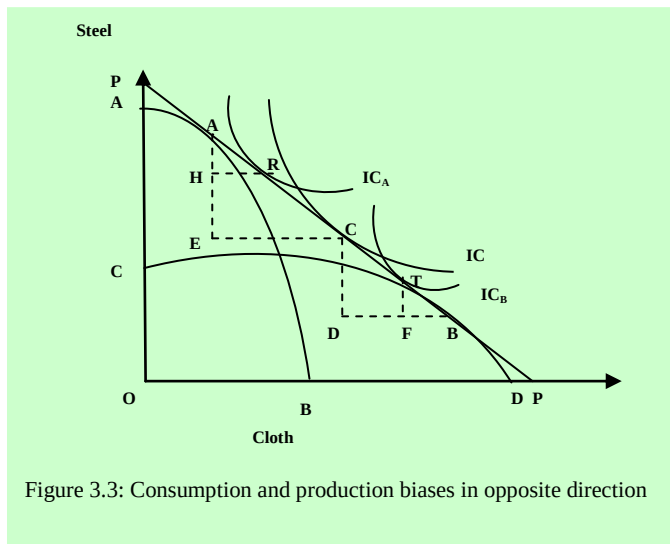
i. Production and Consumption biases in opposite direction

As depicted in figure 3.3 below, after the establishment of trade between the two countries, country A's production shifts to point A (towards greater production of steel), and country B's production shifts to point B (towards greater production of more cloth). This means that the capital surplus country (country A) specializes in the production of the capital-intensive good (steel), and the labour surplus country (country B) specializes in the production of the labor-intensive commodity (cloth). There is greater degree of specialization but by no means complete specialization, in the two countries, because of the diminishing return to scale conditions in the two countries in respect of both the goods.

Figure 3.3 shows that the line PP stands for the international terms of trade line, which is also the relative factor price ratio line after trade is established between the two countries.

(Note incidentally that factor prices will be equalized as a result, of trade). We shall discuss this later under factor price equalization theorem, which is the second proposition of the Modern theory of international trade).

If the demand biases in the two countries are in opposite direction such that we have an indifference curve like IC in figure 3.3, the H-O theorem and its prediction about the structure of trade will hold good. The two countries produce at points A and B but consume at point C. It is important that consumption point, 'such as point C, must lie to the right of point A but to the left of point B (implying opposite direction in production and consumption biases in each country) in order for H-O prediction to be valid. In this case, it has happened in figure 3.3. Country A exports an amount of steel equal to the size AE and imports cloth equal to the size EC.



Country B exports DB units of cloth and imports CD units of steel. The capital surplus country, therefore, is exporting capital intensive good and it is importing labor-intensive good. Similarly, the labor surplus country is exporting labor-intensive good and it is importing capital-intensive good.

In this case, therefore, the H-O prediction would be valid. We emphasize that it is important that consumption should take place to the right of where the production is taking place in country A, and to the left of where the production is taking place in country B. Only then the two countries will specialize in the production as well as export of the commodities which involve intensive use of their respectively abundant factors of production.

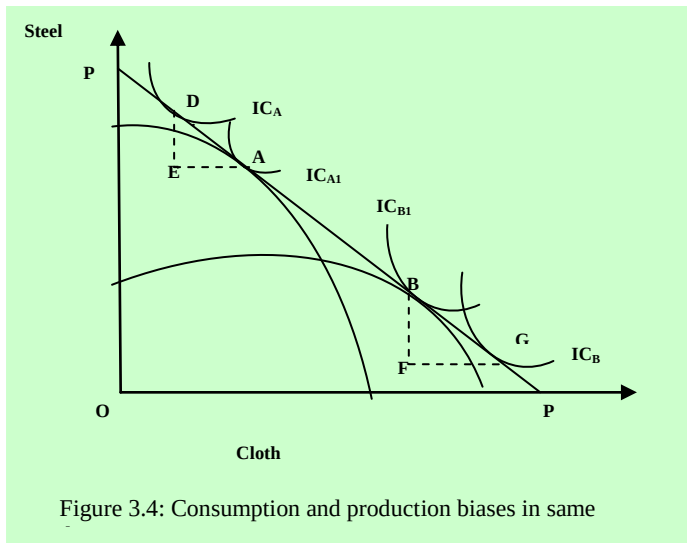
It is by no means necessary that the taste pattern in the two countries must be identical. In figure 3.3, we have made that assumption in drawing a common indifference curve IC both for country A and country B, but it is not necessary. One can feel free to assume that taste patterns in the two countries are different, if that sounds more realistic. For instance, in the same figure, we have also drawn IC_A and IC_B which represent different demand (or utility) patterns in country A and country B. This would only mean that country A is consuming at point R while it produces at point A, and that country B is producing at point B and consuming at point T. Nevertheless, country A exports steel (equal to AH amount) and imports cloth (equal to HR amount); and country B exports FB amount of cloth and imports TF amount of steel. Therefore, as long as the consumption points lie to the right of where production is taking place in country A and to the left of where the production is taking place in country B, the H-O prediction concerning production specialization as well as commodity composition of exports and imports by countries would perfectly hold good.

ii. Production and consumption biases in same direction

This case is illustrated in figure 3.4. The figure reproduces the same information as in figure 3.3, except that in figure 3.4, the demand in country A is biased toward the capital-intensive good and that in country B the demand is biased toward the labor-intensive good. Therefore, as a result, country A produces at point A, specializing in the production of steel.

It consumes at point D, given the utility pattern represented by indifference curve IC_A . This means that country A exports EA amount of cloth and imports ED amount of steel.

Therefore, country A which is a capital surplus country is exporting labor-intensive good (cloth) and importing capital-intensive good (steel). This is in direct conflict with the H-O prediction concerning the commodity structure of trade. Likewise, country B specializes in the production of cloth; it produces at point B. But it consumes at point G in response to its utility pattern represented by the indifference curve IC_B .



Therefore, it exports BF amount of steel and imports FG amount of cloth. Once again we notice that country B, which is a labor-surplus country exports capital-intensive good (steel) and imports labor-intensive good (cloth). The H-O prediction is overturned.

iii. Autarky Situation

If, on the other hand, the demand patterns are not identical that the indifference curve of country A is tangent at point A (as shown by IC_{A1} in figure 3.4) and the indifference curve of country B is tangent at point B (also shown by IC_{B1} in figure 3.4), then it would mean that country A and B choose to consume where they produce. There will then be no trade, but a situation of autarky. In such an event, the H-O prediction will still be valid but, only insofar as it related to production specialization but not structure of trade. There, will, in fact, be no trade to speak of.

3.1.2 Critical evaluation of the H-O theorem

Although the factor proportions theorem developed by Heckscher and Ohlin provides a thorough and plausible explanation of international trade as compared with the classical comparative advantage model, yet it is not free from criticism. The H-O theorem has been criticized mainly along the following three lines of arguments.

- Factor intensity reversal argument
- Leontief's Paradox, i.e. the results obtained by empirical tests conducted by Leontief and others on the capital to labor ratios of exports and imports of developed countries like USA and Japan.

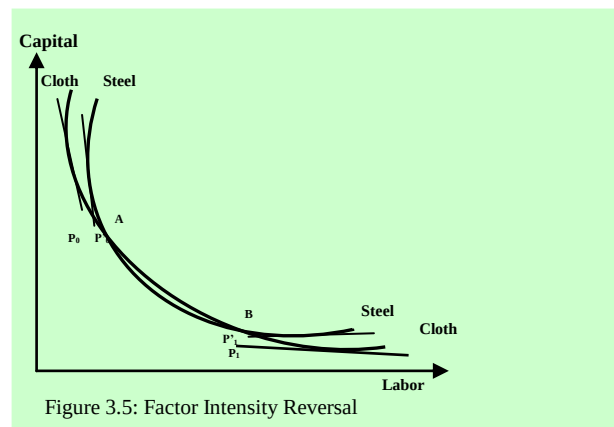
- Demand reversal argument. We have already seen how the H-O theorem will turn out-to be invalid when the demand reversal takes place.

i. Factor intensity reversal argument

The H-O theorem was based on the assumption of no factor intensity reversal. That is, the production functions are different for different goods but they are identical for each good in the two countries. This, in other words, means that one good is capital intensive (with higher capital-labor ratio) and the other good is labor-intensive (with lower capital-labor ratio); but the same good, which is capital-intensive in one country, must be capital intensive in the other country also, and the labor intensive good remains labor intensive in both countries. This assumption is guaranteed when the two production isoquants of the capital-intensive and the labor intensive goods-cut each other only once but not more than once. In figure 3.1, this is shown to happen at point Q. The demonstration in this figure is consistent with the H-O assumption of non-reversibility of factor intensities.

If factor intensity reversal takes place, then the two isoquants would cut each other at more than once and the H-O theorem would turn out to be invalid. This case is demonstrated in figure 3.5. The two production isoquants for steel and cloth cut each other twice in figure 3.5 once at point A and the second time at point B. The factor price ratios in country A (capital surplus country) are represented by the parallel lines P_0 and

P'_0 whereas P_1 and P'_1 represent the factor price ratios in country B (labor surplus country).



In figure 3.5, note the following factors.

- In country A, steel is labour-intensive and cloth is capital intensive. In order to produce one unit of either steel or cloth, country A has to use the same amount of capital but more labor for steel than for cloth. Cloth has a higher capital labor ratio and steel has a lower capital labor ratio. Therefore, a capital rich country like country A would specialize in the production and exports of the capital intensive good, which is cloth. It would import steel which is a labor intensive good.

- In country B, cloth is a labor intensive good and steel is a capital intensive good. Because, to produce one unit of cloth, it takes a given amount of labor and smaller amount of capital as compared to steel. Steel takes the same amount of labor but more capital per unit of output. In country B, therefore, steel has a higher capital labor ratio than in cloth. Naturally, country B (which is a labor surplus country) would choose to specialize in the production and exports of the labor-intensive good, viz. cloth. Country B, therefore, would export cloth and import steel which is capital intensive.

In this case of factor intensity reversal, as we saw above, both the countries produce and export the same commodity i.e. cloth. In the capital rich country, (country A) it is a capital-intensive product, and in the labor rich country, (country B) it is a labor-intensive product. That means the same product (viz. cloth) is capital-intensive in one country but labor-intensive in the other. The same thing applies to steel as well. Steel is a labor intensive product in the capital rich country (country A), and it is a capital-intensive product in the labor rich country (country B). This is a situation of factor intensity reversal. When this takes place, both countries end up producing and exporting the same commodities (cloth) and importing the other commodity steel. This would invalidate the H-O prediction regarding the structure of commodity trade.

ii. Leontief paradox

The first comprehensive and detailed examination of the H-O theorem was the one undertaken by Leontief. You will recall that the theory of factor proportions predicted that the capital abundant country exported capital-intensive goods and imported labor-intensive goods, and the labor surplus country did the opposite. It is commonly agreed that the United States is a capital rich and labor scarce country. Therefore, one would expect exports to consist of capital-intensive goods and imports to consist of labor-intensive goods. Leontief made an extensive study of the US structure of trade and the results were startling. Contrary to what the H-O theory had predicted, Leontiefs study showed that the US exports consisted of labor-intensive goods and the imports, (or more precisely import competing products) consisted of capital-intensive goods. In Leontiefs own words, "America's participation in division of labor in international trade is based on its specialization in labor intensive rather than capital-intensive lines of production. In other words, the country resorts to foreign trade in order to

economize its capital and dispose of its surplus labor, rather than vice versa”. Leontief’s findings are summarized in the following table:

	Exports	Imports Replacements
Capital (US \$ in 1947 prices)	2550780	3091339
Labor (man years)	180313	170004
Capital Labor ratio (US \$ per man hour)	13911	18185

From the, above table, it is obvious that the US exports had a lower capital-labor ratio than the import replacements. Note carefully that these are import replacements produced in the United States as opposed to the actually imported goods in that country.

iii. Demand reversal argument

As we have discussed in figure 3.4 above, if production and consumption biases operate in the same direction in each country, a capital surplus country will go for specializing in the production of capital intensive good but it will export a labor intensive good and a labor surplus country will go for exporting capital intensive good. This structure of trade is contrary to the H-O prediction. Therefore, if demand reversal takes place in both countries, the H-O prediction about the structure of trade will be invalid.

3.2 The factor-price equalization theorem

The factor-price equalization theorem states that the effect of trade is to equalize factor prices between countries, thus serving as substitute for international factor mobility. This theorem has enjoyed far less limelight than the H-O theorem. Nevertheless, it has attracted considerable attention from well-known economists. Heckscher (1919) stated that free trade equalizes factor rewards completely. Ohlin (1933), on the other hand argued that full factor-price equalization cannot occur in practice. Ohlin asserted that free trade brings about only a tendency towards factor-price equalization, and only partial factor-price equalization is possible. The later models by Stolper and Samuelson (1941) and Uzawa (1959) also support partial equalization thesis.

The later works of Samuelson (1948, 1949, and 1953) and of Lerner (1953) make out a case for complete factor-price equalization.

Assumptions of the theorem

In order to demonstrate how the factor-price equalization takes place as a result of international trade, we will use:

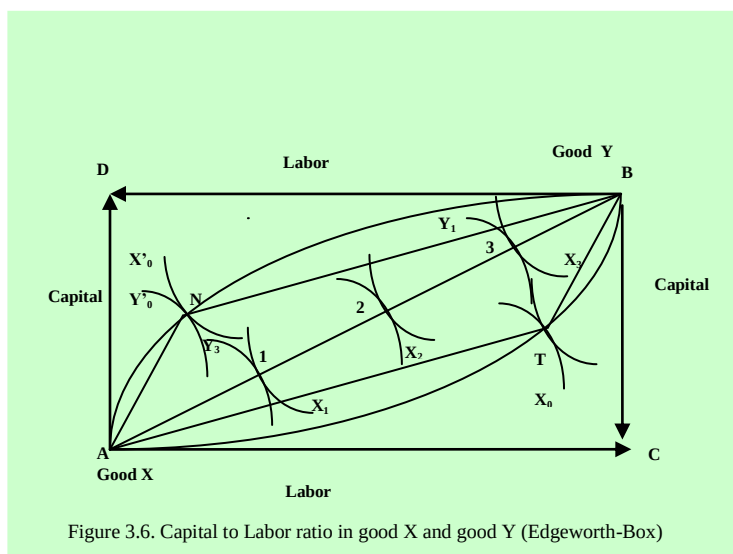
- A model of two countries (country A and B),
- Both countries produce commodities (goods X and Y) and
- There are two factors of production (capital K, and Labor, L) used in each country to produce commodities X and Y.
- Before trade (i.e. in a situation of autarky) we have the following situations:
 - In country A, a labor surplus country, labor is abundant and cheap and capital is scarce and expensive. Therefore, the K/L (or capital labor ratio) is rather low. And once the trade is opened up, labor becomes relatively scarce and the price of labor will go up. Similarly, capital becomes relatively abundant and hence the cost of capital will go down. This follows directly from the H-O theorem that the labor surplus country will specialize in the production and exports of labor-intensive goods. In other words, the abundant factor becomes scarce and the scarce factor becomes abundant, relatively so that (after trade) the K/L will go up in country A.
 - In the capital surplus country-country B-the pre-trade situation is such that capital is abundant and cheap, and labor is scarce and therefore expensive. The K/L will be high before trade. But after trade, this country will specialize in the production of capital-intensive goods, so that the demand for capital will rise relative to that of labor. This will result in capital becoming scarce and the cost of capital going up, and labor becoming abundant and the cost of labor going down. The K/L will drop as a result.

In brief, we start with a low capital-labour ratio in country A and a high capital-labor ratio in country B. This is before trade. But after trade capital ratio will rise in country A and fall in country B until the capital ratios are equalized in the two countries. This is the process by

which the factor prices (capital-labor ratios) in the two countries are equalized as a result of trade. Note that this factor-price equalization is brought about without the movement of factors of production between the two countries. What brings about this factor price equalization, then, is the international trade mechanism. It is in this sense that we can argue that international trade in goods and services, is a substitute for international labor and capital movements (or factor mobility).

We shall now explore the meaning and process of how this factor-price equalization will, in fact, be brought about. We will do it with the help of Edgeworth-Bowley box diagrams.

The points of origin for Good X and Good Y are as shown in figure 3.6. Capital and labor are measured along the horizontal and vertical sides of the above box diagram. The base of the box is larger than the height representing a labor surplus country since it has more supply of labor than of capital.



There are three possibilities with regard to the capital-labor ratio (K/L) in the production of good X and Good Y:

- A Straight Line Contract Curve: If the optimum-efficiency locus (or the contract curve) is a linear straight line, such as AB, the capital-labor ratio in the two goods will be equal and remain so regardless of whether more of X is produced or more of Y is produced. In other words, whether we produce at point 1 or 2 or 3 on the line AB, the capital-labor ratio in good X will be equal to the capital-labour ratio in good Y, which is shown by the equality of the angles $\hat{BAC}$ and $\hat{ABD}$.

If the contract curve is straightline, $\frac{K}{L}$ in good X = $\frac{K}{L}$ in good Y

$\tan \hat{BAC} = \tan \hat{ABD}$, where $\tan \hat{BAC} = \frac{K}{L}$ in good X and $\tan \hat{ABD} = \frac{K}{L}$ in good Y

- A non-linear contract curve representing difference in the capital to labor ratios of good X and Y.
- If the contract curve is non-linear of the type represented by the line ANB in figure 3.6), Good X will be a capital-intensive good and Good Y will be a labor-intensive good.

Because, $\frac{K}{L}$ in good X > $\frac{K}{L}$ in good Y

$\tan \hat{NAC} > \tan \hat{NBD}$, where $\tan \hat{NAC} = \frac{K}{L}$ in good X and $\tan \hat{NBD} = \frac{K}{L}$ in good Y

- Finally, if the contract curve is non-linearly shaped like the sagging line ATB in figure 3.6, we find that good X is labor-intensive and good Y is capital-intensive.

Because, $\frac{K}{L}$ in good X < $\frac{K}{L}$ in good Y

$\tan \hat{TAC} < \tan \hat{TBD}$, where $\tan \hat{TAC} = \frac{K}{L}$ in good X and $\tan \hat{TBD} = \frac{K}{L}$ in good Y

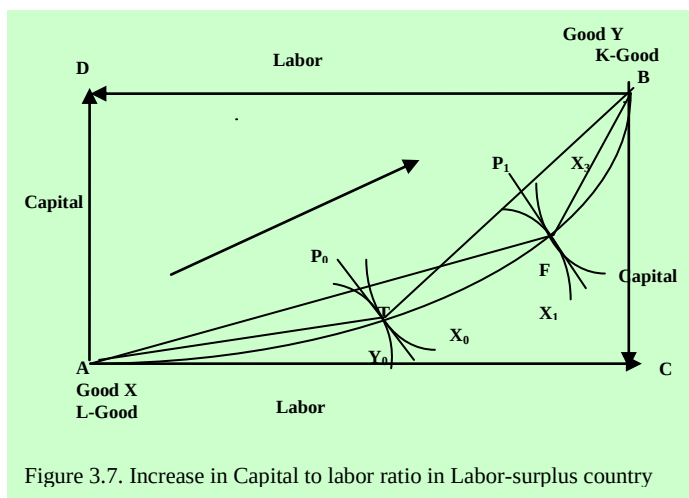
Throughout what follows, we will assume that the contract curve is of the type represented by the non-linear line ATB so that good X is labor intensive and good Y is capital intensive. Even if the country had more capital and less labor, good X will remain labor-intensive and good Y capital-intensive, so long as we have a contract curve of the shape of ATB as shown in figure 3.6. This would mean that in our model:

- country A is a labor surplus country and country B is capital-surplus country,
- Good X is labor-intensive and good Y is capital-intensive in both the countries, regardless of the differences in factor proportions and factor prices in the two countries.

We shall now turn to the process of factor-price equalization between the two countries, as a result of the opening of trade between them. We will first show what happens in the two countries, using separate graphs, and later on put them together in one composite graph.

3.2.1 Labor Surplus Country Case

The Box ACBD shows total factor supplies in a labor surplus country A. The contract curve has the shape of ATB. The production isoquants X_0 and X_1 are for good X; and Y_0 and Y_1 are for good Y. Before trade, the country produces at point T, where X_0 is tangent to Y_0 at factor price ratio represented by the line P_0 . At point T the capital-labor ratio in good X is equal to $\tan \hat{TAC}$, and the capital-labor ratio in good Y is equal to $\tan \hat{TBD}$. The size of angle TBD is more than the size of TAC, which shows that good Y is capital-intensive and good X is labor intensive in country A.



Once trade is opened, this country will go for specializing in the production (and exports) of good X. This would result in a rightward shift along the contract curve from, say, point T to point F. In moving from point T to F, there is an increase in the production of X and a decrease in the production of Y, which is indicated by an upward movement of the X isoquant and a downward movement of the Y isoquant. In moving from point T to F, there has been an increase in the capital-labor ratio in X production (from $\hat{TAC}$ to $\hat{FAC}$) as well as Y production (from $\hat{TBD}$ to $\hat{FBD}$). You will, therefore, notice that in country A (a labor surplus country) the capital-labor ratio in both goods production has gone up after the establishment of trade. This is indicated by the change in the slope of factor price line (slope of $P_0 < \text{Slope of } P_1$). This change in slope of the price line indicates that the factor price ratio in terms of P_L/P_K has increased as we move from point T to point F.

3.2.2 Capital Surplus Country Case

The box ACBD in figure 3.7 represents factor supplies in country B, a capital surplus country. Before trade country B produces at point T. At this point, the capital-labor ratio in the production of X was equal to the angle TAC, and in the production of Y it was equal to TBD. Trade results in specialization towards increased production of Y (the capital intensive good), and therefore, the point of production shifts from point T to, say, point F. As

a result of this production-shift, we find that the capital-labor ratios in the production of both X and Y has decreased in this country.

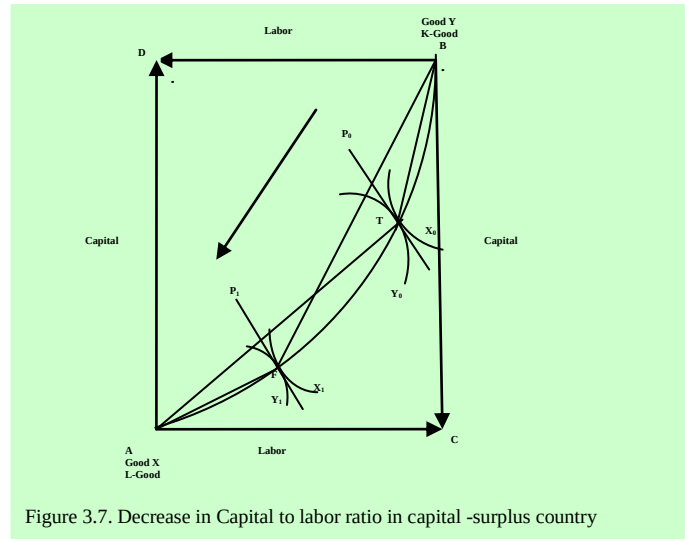


Figure 3.7. Decrease in Capital to labor ratio in capital-surplus country

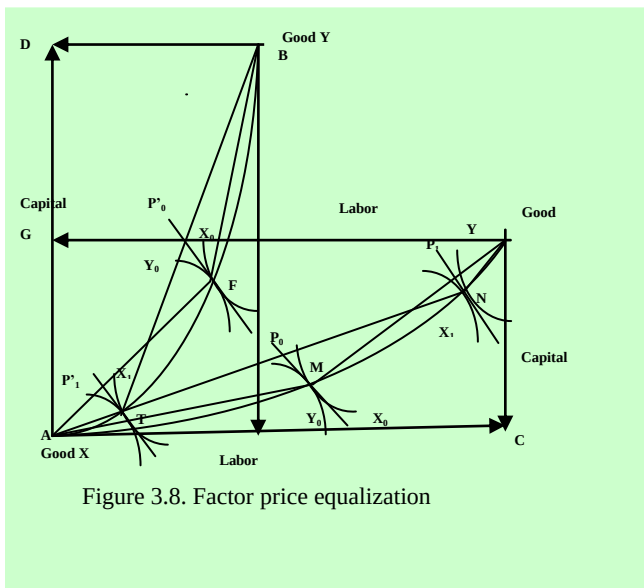
In respect of X, the capital-labor ratio has decreased from $\tan \hat{TAC}$ to $\tan \hat{FAC}$; and in respect of Y; it has decreased from $\tan \hat{TBD}$ to $\tan \hat{FBD}$.

The change in capital-labor ratios in the production of both X and Y is indicated by the change in the slope of the relative factor-price ratio line from P_0 to P_1 where slope of $P_0 >$ slope of P_1 .

Thus, in the capital surplus country-country B, the capital to labor ratio has decreased in the production of both X and Y goods, as a result of trade. When this is combined with the increase in capital-labor ratio in the production of the two goods in country A, a labor surplus country, we eventually get to a point where the capital-labor ratios (or factor prices) are equalized in the two countries.

3.2.3 Factor price equalization: a composite graph

Figure 3.8 is a joint of figures 3.6 and 3.7. Under autarky, country A produced at point M and country B at point R. The two countries' factor price ratios were different, as measured by the slope of the line P_0 and P'_0 in the two countries. After the establishment of trade between the two countries, there is a shift in the equilibrium production points. The equilibrium production point shifts from point M to N in country A, and from point F to T in country B. P_1 factor price ratio line in country A (at point N) has the same slope as the P'_1 line for country B (at point T), which shows that trade has led to equalization of factor prices in country A and B. The slope of the price lines are the ratio of price of labor to price of capital in each country.



Thus, we see that international trade brings about equalization of factor prices between countries even in the absence of factor movements between the countries. It is in this sense that we argue that international trade in goods and services, is a substitute for international movement of labor and capital. In other words, international trade brings about equalization of both the product prices and the factor prices.

3.3 Critical evaluation of the factor price equalization theorem

The factor price equalization theorem is based on certain assumptions which are rather unwarranted in real life. The following key points are worth taking note of, because they constitute obstacles to the equalization of factor prices in the real world.

1. The theorem assumes complete free trade, i.e. the absence of tariff and non-tariff barriers to trade. It also assumes that there are no transport costs in exporting and importing goods and services between nations. In real world, we know that both exist. Their very existence, therefore, will prevent factor price equalization that free trade could have otherwise brought about. Complete equalization of factor prices could take place if, and only if, the factors of production were internationally perfectly mobile. In that sense, international trade in goods and services is not perfect substitute for international factor mobility. It is at best a close substitute.
2. International trade would only lead to partial or incomplete specialization, but by no means to complete specialization in production. This will, therefore, rule out the possibility of complete factor price equalization. At best, we can expect partial equalization of factor prices. This conclusion is based on the premise that there are diminishing returns to scale conditions in the production of all goods in all countries.
3. The theorem is based on the assumption of perfect competition and diminishing- returns to scale in production. In the real world, however, there is imperfect or monopolistic competition on the one hand, and on the other hand, there are increasing returns to scale in the production of some goods. This would destroy the credibility of the factor price equalization theorem.
4. For complete factor price equalization to take place the number of factors should not exceed the number of products. In a model of two countries, two factors and two goods, it is possible to show factor price equalization. But, in a real world model, of many countries, many factors and many goods and services, it is not possible to argue for a complete equalization of factor prices.
5. The theorem would collapse once we show that the production functions are not identical in all the countries taking part in international trade. The theorem will not hold good if the factor intensity reversal takes place, because in that event a capital rich country and a labor rich country will export the same good by using different techniques of production suited to their factor endowments. Factor intensity reversal creates obstacle to factor price equalization.
6. The theorem assumes that factor supplies remain fixed in every country. This is unrealistic, because we do know that the supplies of labor and capital keep constantly changing over time, almost in every country.

CHAPTER SUMMARY

- The modern theory of international trade is based on two propositions: the factor endowment theory (H-O theorem) and the factor price equalization theorem.
- The factor endowment theory states that a country has to specialize in the production and export of the good that uses most the country's relatively abundant (cheap) factor of production and import the good that uses most its relatively scarce (expensive) factor of production.
- The H-O prediction about the structure of trade between countries that differ in factor endowment holds true under the price criterion of defining factor abundance whereas it may or may not hold true under the physical criterion. Under the physical criterion it holds true if production and consumption biases in each country are in opposite direction and will not hold true otherwise (i.e. if demand reversal takes place).
- The H-O model is criticized based on three main lines of arguments called the factor intensity reversal argument, the demand reversal argument and the empirical works by Leontief and his associates (i.e., Leontief's paradox).
- According to the factor price equalization theorem, the objective of international trade in goods and services is to equalize factor price differences between countries and hence serve as a substitute for international factor mobility.
- In an Edgeworth-Box, a straight line contract curve represents no factor intensity difference between two commodities in production whereas a non-linear contract curve below and above the main diagonal of the box represent difference in factor intensity between the two commodities (i.e., if one is relatively capital intensive the other good will be labor intensive).
- A contract curve is a line that joins the locus of points at which the isoquant curves of two commodities are tangent to each other. At the tangency point, the slopes of the two contract curves are equal representing that the marginal rate of technical substitution of labor for capital for the two goods are equal.

REVIEW QUESTIONS

1. State the two main propositions of the modern theory of international trade.
2. Explain the two alternative criteria used to define factor abundance in the H-O theorem.
3. What does that H-O theorem predicts about the structure of trade between countries with different factor endowments?
4. Explain H-O model under the price criterion using graphs and the basic assumptions.
5. Analyze the situation of demand reversal under the physical criterion of defining factor abundance with the help of graph.
6. Explain graphically the factor intensity reversal argument.
7. Define the contract curve.
8. Show the three possibilities with regard to the capital-to-labor ratio in producing good X and good Y using the Edgeworth-Box.
9. Explain how factor price equalization takes place between countries with different factor endowments (one country as labor surplus and the other as capital surplus country) using composite graph of the Edgeworth-Box.
10. Write the basic criticisms of the factor price equalization theorem.

CHAPTER FOUR**4. INTERNATIONAL TRADE POLICY****CHAPTER OBJECTIVES**

The objective of this chapter is to acquaint students with theories of international trade policy and their impact on trading countries. After learning the contents in this chapter, students will be able to:

- *Explain the arguments for and against free trade policy*
- *Explain the arguments against protectionist policy of international trade.*
- *Identify and explain the tariff and non-tariff barriers to free trade.*
- *Identify the types of import and export tariffs on goods and services.*
- *Explain graphically the price, consumption, production (import substitution), revenue, balance of payment, welfare, terms of trade, and employment effects of import tariffs on the tariff imposing country*
- *Explain why nations often prefer non-tariff barriers to tariff barriers as international trade policy.*
- *Identify the types of quota; explain the effects of quota on quota imposing country.*
- *Explain the differences and similarities between import quota and import tariffs*
- *Explain the effects of export subsidies on trading countries.*

4.1 Free trade: Case for and against

A policy of no restrictions on the movement of goods between countries is known as the policy of Free Trade. Restrictions placed with a view to safeguarding home industries constitute the policy of protection. In the words of Adam Smith, the term 'free trade' has been used to denote "that system of commercial policy which draws no distinction between domestic and foreign commodities and, therefore, neither imposes additional burdens on the latter, nor grants any special favors to the former." Free trade, however, does not require the removal of all duties on commodities. It only insists that they shall be imposed exclusively for revenue and not at all for protection.

Adam Smith wrote: "If a foreign country can supply us with a commodity cheaper than we ourselves can produce, better buy it from them with some part of the produce of our own

industry, employed in a way in which we have some advantage." Be continued further: "Whether the advantage which one country has over another be natural or acquired is in this respect of no consequence. As long as one country has those advantages and the other wants them, it will always be more advantageous for the latter rather to buy of the former than to make." The only exception that Adam Smith would make was industries necessary for defense. These might be protected since defense is more important than opulence, he said.

The doctrine of free trade is the extension of the doctrine of division of labor to the international field. In the words of Adam Smith again, "Individuals find it for their interest to employ their industry in a way in which they have some advantage over their neighbors." And he adds, "What is prudence in the conduct of every private family can scarce be folly in that of great kingdom." In short, the free trade theory is that such a policy enables every country to devote itself to those forms of production for which it is best suited on the basis of comparative advantages.

4.1.1 Cases for free trade

For arguments in favor of free trade see advantages of foreign trade discussed in chapter I.

- Gains from foreign trade:
 - Static gains = to welfare gains in consumption and production
 - Dynamic gains impact on economic growth and economic development

4.1.2 Cases against free trade: protectionism

The term 'protection' is used to denote a policy of encouraging the home industries by the use of bounties or by the imposition of high customs duties on foreign products. The objective is to build up great national industries even by sacrificing utilities on the part of existing consumers. The arguments for protection are stated below.

1. Infant Industry Argument.

This argument was put forward by Friedrich List, J. S. Mill, Alfred Marshal and Souderton.

Friedrich List

Friedrich List was a German economist, in 1840. In those days, Germany was making all-out efforts to industrialize its economy. On the other hand, the U.K. had already built by that

time, a sound industrial base. Germany was then facing lot of difficulties in competing with the established British industry. List had strong view that the development of industries is a pre-requisite of economic progress. Free trade was good for Britain whose economic position was already well established. For young emerging German industry, however, protection of a tariff wall was essential.

J. S. Mill

In the words of Mill, "The superiority of one country over another in a branch of production often arises only from having started it sooner. There may be no inherent advantage on one part or disadvantage on the other, but only a present- superiority of acquired skill and experience. Thus, J. S. Mill, one of the advocates of free trade, accepted only one argument. In Mill's words it ran thus: "A productive duty, continued for a reasonable time, might sometimes be the least inconvenient mode in which a nation can tax itself for the support of such an experiment (introducing new industries). But it is essential that protection should be confined to cases in which there is good ground of assurance that the industry which it fosters will after a time be able to dispense with it.

Alfred Marshall

This prominent English economist, too, conceded the force of the infant industry argument, although the English economists by and large advocated free trade. However, for protection to produce social benefits, an infant industry must first grow up. It must eventually be able to compete at world market prices. Not only has it to grow up, for a protectionist policy to be profitable, according to Sodersten, it will have to be able to pay back the losses due to protection during the infant industry period. Only then is there a clear-cut case for infant industry protection.

This argument specially applies to countries that enter the industrial field at a later stage. Such countries possess potential advantages which may not become effective unless- foreign competition is excluded for a period of time. Unless an industry in its infancy is protected till it acquires strength and maturity, it will die in the face of foreign competition.

The infant industry argument rests on the following grounds:

- Every country has its own potentialities for developing some industries in the form of labour skill, raw materials and entrepreneurial talent. No country is completely devoid of productive elements which may yet be latent.
- Every nation has the right to develop its potentialities or discover its latent productive powers. No nation would like its efforts for economic development to be smothered before it has had a full chance for trial and error.
- It takes time for the productive elements to be developed. Labour can be trained; raw materials can be improved; and entrepreneurs can become competent by experience. But they should have enough time to develop and discover themselves.
- Infant industries cannot be expected to withstand competition from old and well-entrenched industries. It will be unfair to expose new industries.
- If infant industries are not duly protected against competition from strong and well-established industries, they are bound to die. This will mean waste of valuable national assets invested in the industry and a serious setback to entrepreneurial venture in future.

In spite of these weaknesses, the argument has been widely accepted and many countries have industrialized themselves through protection given on the basis of this argument, *e.g.*, the U.S.A. and several members of the British Commonwealth, including India. But it is recognized that protection should not be given on a permanent basis. It should be given for a definite period considered sufficient for the industry to grow. Moreover, it should not be given indiscriminately to all industries. The industries to be given protection should be selected with proper care and discrimination so that scarce productive resources of the community are properly allocated.

2. Diversification of Industry Argument

This argument was advanced, among other writers, by Friedrich List in Germany. According to List, a nation should have a variety of sources of production and employment. Depending on one industry or on a few industries is dangerous both politically and, economically. Politically it means too much dependence on foreign trade which may be cut off during a war. Economically, a country depending on a few industries is exposed to the danger of serious economic dislocation in case some adverse circumstances affect such an industry.

But it must be understood that this argument cuts at the root of the principle of comparative cost according to which each country must specialize in the production of certain articles. According to this argument a country must produce even those articles in which it may not have comparative advantage.

3. Employment Argument

It is argued that industrial development through protection increases employment in a country. Conversely, if protection is not given to old established industries; foreign competition may ruin them and create unemployment in the country.

4. Conservation of National Resources Argument

Carey and Patten had argued that free trade resulted in the export of agricultural commodities from America and thus led to the exhaustion of the soil. Jevons in England applied the same argument again", the export of coal which exhausted coal fields. The same argument has also been applied the Union of South Africa regarding gold mining and in India regarding the export of manganese.

The argument has some force. If a country exports its exhaustible materials in a raw-state, it loses manufacturer's profits. It may also be seriously handicapped when such materials have been altogether exhausted.

5. Defense Argument

Adam Smith remarked "Defense is better than opulence". It is said that essential to make a country militarily strong though it may not be economically prosperous. Hitler preached to the German nation, "Gun! Better than butter." According to this argument a country must actively encourage the development of those industries which are essential from the point of view of defense, even though it may result uneconomic distribution of the national resource. The advocates of free trade point out that this is politics and not economics. On purely economic grounds, they say, free trade is the best.

6. Revenue Argument

Protection is also advocated for revenue purposes. When protective import duties are imposed, they certainly bring in revenue to the government. But it may be pointed out that there is a certain degree of incompatibility between the revenue and protection. If full

protection is given, the government will not get any revenue, because full protection will mean that domestic goods have driven foreign goods altogether. When foreign goods not come in, there will be no revenue from import duty. On the other hand, if government wants revenue then foreign goods must come in and compete with domestic goods. Then domestic industries do not get any protection. This incompatibility, however, arises between maximum protection and maximum revenue. But if the duties are moderate, they will yield revenue besides affording protection. It is, however, much better to advocate protection for the sake of protecting industries rather than for raising revenues.

7. Key Industry Argument

If the industrial structure of a country is to be stable and sound, the country must develop 'key' or basic industries; otherwise foundation of industries will have been laid on sand. The country may not have any comparative advantage in such industries. But since they are of crucial importance and have to be developed, protection must be granted to them.

8. Balance of Payments Argument

It may become necessary to check imports by means of tariff in order to rectify adverse balance payments. The I.M.F. regulations permit member countries to impose temporary restriction trade to cure a balance of payments deficit. Import restrictions on non -essential imports become necessary in the interest of accelerated economic growth.

9. Patriotism Argument.

Protection is advocated on patriotic grounds also. It is the duty of every citizen to use home-made goods as far as possible. W-f: must, therefore, develop our industries, through protection, if necessary, so that home-made goods in the right quantity and of good quality are made available for use.

10. Self-sufficiency Argument

Another argument in favor of protection is that we should become self-sufficient and not depend on other countries for our necessities. Such dependence proves very dangerous during war when foreign trade is cut off.

11.Avoid unfair competition

Protection also becomes necessary against unfair competition from abroad arising from dumping, depreciated exchanges, bounties, *etc.*

12.For Economic Stability

Protection is also advocated to shut out the baneful influences trade cycles from abroad. It is expected to make the domestic economy immune from the destabilizing effects of external disturbing factors. In the Macmillan Committee (1931), the late Lord Keynes put forward the opinion that protection and not free trade was needed to restore the much-needed economic stability for an economy which is out of gear.

4.1.3 Arguments against protection

1. **Vested interests are created:** Once certain industries are given protection, they claim it as a matter of right. It then becomes very difficult to take away protection. The infants begin kicking if you touch them in any manner. Such infants refuse to admit that they have grown into adults.
2. **Protection produces lethargy and acts like an opiate:** When foreign competition has been removed, it sends the home manufacturers to sleep, as it were. They do not try to make any improvement, and technical progress comes to a standstill.
3. **Then there is the danger of corruption:** The industrialists bribe legislators so that protection is not taken away. This evil was rampant in the U.S.A. at one time.
4. **Protection creates monopolies:** Tariff is said to be the mother of trusts. When foreign competition has been removed, the home manufacturers are tempted to combine to reap monopoly profits.
5. **Consumers and unprotected industries suffer:** This is so because imposition of import duties invariably leads to the rising of prices.
6. **The distribution of wealth becomes more unequal:** Protection favors the rich capitalists who grow still richer. The gulf between the 'haves' and 'have-nots' is thus widened still further.
7. **Source of conflict:** Protection leads to conflict, friction and retaliation in international dealings. It thus breeds the germs of future wars.
8. **Inefficiency in world resource allocation:** The most important argument against

protection on economic grounds is that it militates against optimum utilization of resources. It hampers international division of labor so that labor, capital and other factors of production do not find their most remunerative employment. Their distribution is not governed by natural economic forces-, but they are artificially forced into certain channels. The result is that they do not make their maximum possible contribution in the production of commodities. The world output is lower than it could be, so that the standard of living is necessarily lowered. A natural movement towards world prosperity is hindered.

4.2 Barriers to Trade

The two broader categories of barrier to free trade between countries are natural barriers and man-made barriers. Natural barriers to trade: arise on account of the cost and the distance involved in moving goods and services from one country to another. In short, it is the transport cost. Man made barriers is further classified into tariff and non-tariff (hidden) barriers.

4.2.1 Tariff and Its Effects

What is tariff?

Tariffs are essentially the taxes or duties imposed on the imported or exported goods. Import duties are more common than export duties, so much so that tariffs are often identified with import duties. Tariffs are aimed at altering the import prices (or export prices) so as to regulate the volume of imports (or of exports).

Types of Tariffs

- 1) Specific tariffs: are tariffs which are assessed on the basis of the physical weight of the product which is imported or exported. The units are in terms of Birr/Kg, Birr/ tone. They can be levied on goods like wheat, sugar, coffee, cattle, etc. Specific tariffs cannot be levied on valuable goods like diamond, modern art paintings, TV etc.
- 2) Ad valorem tariffs: are tariffs levied based on the value of the product. It is a percentage tax. E.g. 10% of the value of diamond, TV etc. Ad valorem tariff is revenue elastic but specific tariff is not.
- 3) Compound tariff: combines a specific duty with an ad valorem duty.
- 4) Discriminatory tariff calls for different rates of duties depending on the country of origin or destination of the product

- 5) Non-discriminatory tariffs: uniform tariffs rates imposed on goods and services regardless of their source of origin or destination. Tariffs are said to be single column when they are non- discriminatory and double-column when they are discriminatory.
- 6) Revenue tariffs: these are tariffs that are imposed primarily and produce revenue for the government
- 7) Protective tariffs: are tariffs that are imposed primarily to protect the domestic industries from foreign competitions
- 8) Retaliatory tariffs: when country A imposes (increases) duties against the products from country B, it is possible that country B will retaliate and levy duties on goods imported from country A. Thus, country B's tariffs are then described as retaliatory tariffs.
- 9) Countervailing tariffs: Tariffs are said to be countervailing when a country imposes (increases) import duties with a view to offset export subsidizing in the country of origin.

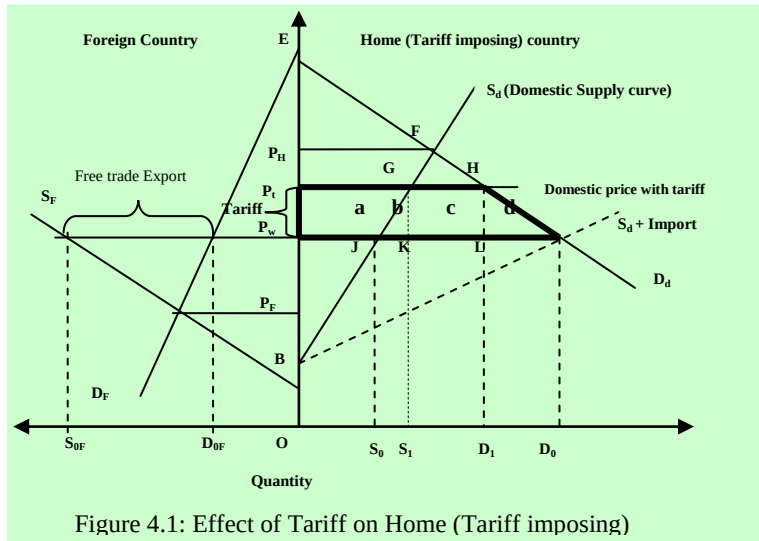
4.2.1.1 Effects of import tariff on tariff imposing country

Tariff is an important tool of commercial policy. Although it is primarily a protectionist device yet it proves to be a double-edged weapon. On the one hand, it limits consumers' choice by forcing them to cut consumption of the goods they like and, on the other; it shifts the use of resources from one use to another. The effect of tariffs is to change relative prices of goods, services and factors of production.

Figure 4.1 shows the foreign country's demand and supply for a given product on the left side of the graph and to the right is the home (tariff imposing) country's demand for and supply of same (homogeneous) product. P_H and P_F are the pre-trade equilibrium prices in the home and foreign countries respectively. The price differential ($P_H - P_F$) induces trade between the two countries letting the home country importer and the foreign country an exporter of the product.

When international price is equalized at P_w (world price), the foreign country's exports ($S_{0F}D_{0F}$) are equal to the home countries imports ($S_0D_0 = JM$). The free trade price is therefore P_w at which the home country's imports are JM . Transport costs and tariffs are assumed to be zero. Now let us introduce tariffs. If the rate of import tariff per unit of imported product is equal to the distance

between P_t and P_w , this imposition of tariff by the home country on the imported product will cause the following effects on the importing country.



1. Price effect

The immediate effect of imposition of tariff is to raise the price of both the imported product and the domestically produced product (import replacement product) in the home country. Figure 4.1 shows that due to tariff the product prices increases from P_w to P_t . Thus, the import duty is paid as well as borne by the importing country exclusively.

2. Consumption effect

When tariff is imposed, price of the commodity rises and domestic consumption is reduced from OD_0 (free trade consumption) to OD_1 (consumption after tariff) following the law of demand. The reduction in consumption due to tariff is equal to $(D_0D_1 = ML)$. This is called the consumption effect.

3. Production (Import substitution), or protective effect

The domestic production will go up as a result of the tariff, except in a limiting case where the domestic supply curve is perfectly inelastic. In figure 4.1, the domestic supply curve S_d is elastic so that when the tariff raises the price from P_w to P_t , the domestic production of import replacement goods goes up from S_0 to S_1 . The amount by which domestic production goes up as a result of tariff can be described as the extent of import substitution. In the figure the distance $S_0S_1 = JK$ is a measure of this import substitution. Tariff reduces the import

competing goods thus affording protection to the domestic producer. Domestic production increased as shown above as a result of the imposition of tariff.

4. Revenue Effect

The government derives revenue from the tariff which is measured by the quantity of the imports multiplied by the rate of tariff. The total tariff revenue that government collected amounts to the size of the rectangle (GHLK = c).

5. Balance of trade effect

This effect of tariff is the result of tariffs on the level of imports. At the free trade price P_w , the level of imports was $S_0D_0 = JM$. The country was paying a foreign currency equivalent to area of the rectangle $S_0JMD_0 = OP_w \times S_0D_0$ as an import payments. But after tariff, the import quantity has been reduced to $S_1D_1 = KL$ and the import payment has also been reduced to the area of the rectangle $S_1KLD_1 = OP_w \times S_1D_1$ from the free trade import payment level. Thus this will lead to improvement in the balance of trade (i.e., either increasing trade surplus or reducing trade deficits) of the tariff imposing country. Out of the total import cuts from S_0D_0 to S_1D_1 , the reduction amounting $S_0S_1 = JK$ is due to production (import substitution or protective) effect of tariff and the other $D_1D_0 = LM$ is due to consumption effect of tariff.

6. Economic welfare effect

The welfare effect of tariffs is, perhaps, the most important effect. The economy of a country is composed of three parties, namely, consumers, producers and the government. In considering the economic welfare effect of tariffs on the imposing country, we need to find out whether the welfare of each of the groups in the economy has gone up or down.

a. welfare effect of tariffs on consumers

Before tariffs were imposed (or at the free trade price P_w), the price paid by the consumers per unit of product was OP_w . At this price consumers buy OD_0 quantity of the product. Since the demand curve Dd is also the marginal utility curve for the product, under free trade equilibrium, consumers can derive a total utility of $OEMD_0$ and to derive this much of utility they need to pay OP_wMD_0 as consumption expenditure. The difference, which is equal to the area of the triangle P_wEM , is the net utility or consumers' surplus that consumers can derive at the free trade world price P_w . This consumers' surplus is the measure of the total welfare of consumers at the free trade price.

Imposing a tariff rate of $P_w P_t$ per unit of imports will reduce the consumers' surplus from the free trade level to an area of the triangle $P_t E H$. This implies that consumers will lose a welfare level equal to the area $P_w P_t H M = a + b + c + d$. Thus, tariffs decrease the welfare level of consumers through increased prices and decrease in quantity demand (consumed).

b. welfare effect of tariff on producers

At the free trade price P_w , the level of producers' surplus that measures of the welfare of domestic producers is equal to the size of the triangle $B P_w J$. It is the difference between revenue (the area $O P_w J S_0$) and cost (the area $O B J S_0$) of domestic producers.

After tariff, the producers' surplus increased to the size of the triangle $B P_t G$. The net gain in producers' surplus equals the area $P_w P_t G J = a$. This gain in producers' surplus is part of the loss in consumers' surplus. Moreover, the gain in producers' surplus is due to the protective (production or import substitution) effect of tariff.

c. welfare effect of tariff on government

Before tariff, there is no revenue to the domestic government collected from import duties. The tariff rate of $P_w P_t$ multiplied by the amount of import quantity $S_1 D_1 = K L$ generates a tariff revenue of the size of the rectangle $K G H L = c$ to the tariff imposing country's government. This revenue is also part of the welfare loss by consumers.

d. Overall welfare effect of tariff

The imposition of tariff increases the price of the commodity and thus reduces the consumers' surplus. In this way income is transferred from the consumers to producers and the government. This effect of distribution of income is called the distributive effect. However, all the welfare loss by consumers is not redistributed to other groups in the society. There is a dead weight loss, also termed as the cost of tariff. This loss equals the area $b + d$. It is the net welfare loss to the economy due to tariff.

7. The Terms of Trade Effect.

Take the case that the foreign and the home countries with different factor endowments giving comparative advantage to each production of a certain commodity. Tariff reduces the volume of trade and the terms of trade will improve for the country imposing the tariff.

8. Effect on National Income

If a country is facing unemployment problem, imposition of tariff increase employment and thus increases income. This happens because with the imposition of tariff consumers' demands are diverted to domestically produced goods. To meet this increased demand for domestic products new production units will be set up. As a result lot of employment will be created and national income increased.

4.2.2 Non-tariff barriers

Non-tariff barriers encompass a variety of measures. Some have insignificant trade consequences; for example, labeling and packaging requirements can restrict trade, but generally only marginally. Other non-tariff barriers significantly affect trade patterns; examples include import quotas, voluntary export restraints, subsidies and domestic content requirements. These non-tariff barriers are intended to reduce imports and benefit domestic producers. There are several reasons suggested for why countries use non-tariff barriers instead of tariff.

1. Political reason

Politicians may sometimes be reluctant to levy a tariff because it is an explicit tax on domestic consumers. Although non-tariff barriers result similar consumers' welfare loss like tariff, the non-tariff barriers' effect tends to be more remote and less visible. Politicians may therefore perceive non-tariff barriers as being more acceptable to the voting public.

2. Certainty of restrictive impact of non-tariff barriers

The restrictive impact of non-tariff barriers is relatively more certain as a protection instrument than tariff. For example it may be much easier to show that a quota of 5 million tons restricts food imports to 5 million tons than to show conclusively that a tariff of Birr 20 per ton would result in imports of only 5 million tons of food imports.

Moreover, a rise in an import tariff may encourage the exporting country to subsidize its exporting companies to make them more price-competitive, thus offsetting the tariff's restrictive impact. Such subsidies, however, would not alter the restrictive impact of an import quota or other non-tariff barriers.

3. Non-tariff barriers are not covered by rule of WTO (former GATT)

Many nations have adopted non-tariff barriers because they are not covered by the main body of rules of conduct in international trade, the General Agreement on Tariffs and Trade (GATT). Although GATT has succeeded in liberalizing tariffs in the post-World War II era, nations have circumvented GATT rules by using loopholes in the agreements and levying types of non-tariff barriers over which trade negotiations have failed.

4.2.2.1 Import quota

A Quota is a direct limitation of the physical quantity of exports and imports permitted in the country. We will only discuss import quotas as they are more common than the export quotas. An import quota is a physical restriction on the quantity of goods that may be imported during a specific time period; the quota generally limits imports to a level below that which would occur under free-trade conditions. The effects of quotas are similar to those of tariffs, but there are also substantive differences between the two which are worth examining.

Types of Quotas**1) Unilateral quotas**

Unilateral quotas can be global and has price advantage, or allocated. Import quotas are generally imposed *unilaterally* by the home government, without prior negotiation or consultation with other nations. They are levied and administered exclusively by the importing nation. Because of their unilateral nature, import quotas may be resented by other nations and may lead to retaliatory quotas.

a. Global quota

One way of administering import limitations is through a global quota. This technique permits a specified number of goods to be imported each year, but does not specify where the product is shipped from or who is permitted to import. When the specified amount has been imported (the quota is filled), additional imports of the product are prevented for the remainder of the year.

In practice, the global quota becomes unwieldy because of the rush of both domestic importers and foreign exporters to get their goods shipped into the country before the quota is filled. Those who import early in the year get their goods; those who import late in the year may not. Moreover, goods shipped from distant locations tend to be discriminated

against because of the longer transportation time. Smaller merchants, without good trade connections, may also be disadvantaged relative to large merchants. Global quotas are thus plagued by accusations of favoritism against merchants fortunate enough to be the first to capture a large portion of the business. For these reasons, global quotas are relatively uncommon, especially among industrial nations.

b. Selective (allocated) quota

To avoid the problems of a global quota system, import quotas are usually allocated to specific countries; this type of quota is known as a selective quota. For example, the Ethiopia might impose a global quota of 10 million tons of canned food per year, of which 3 million must come from the Kenya, 4 million from Germany, and 3 million from Saudi Arabia. Customs officials in the importing nation monitor the quantity of a particular good that enters the country from each source; once the quota for that source has been filled, no more goods are permitted to be imported.

2. Bilateral quotas

Bilateral quotas imply mutual agreement between countries through negotiations. They do not provoke retaliation.

3. Mixing or indirect quotas

In this case the domestic producers are asked to use a fixed proportion of imported and domestic materials used in producing their products. The quota is fixed not in absolute forms but in percentage forms.

4.2.1.1 Tariffs versus quotas

The effects of quotas are similar to those of tariffs, but there are also substantive differences between the two which are worth examining.

a. Similarities

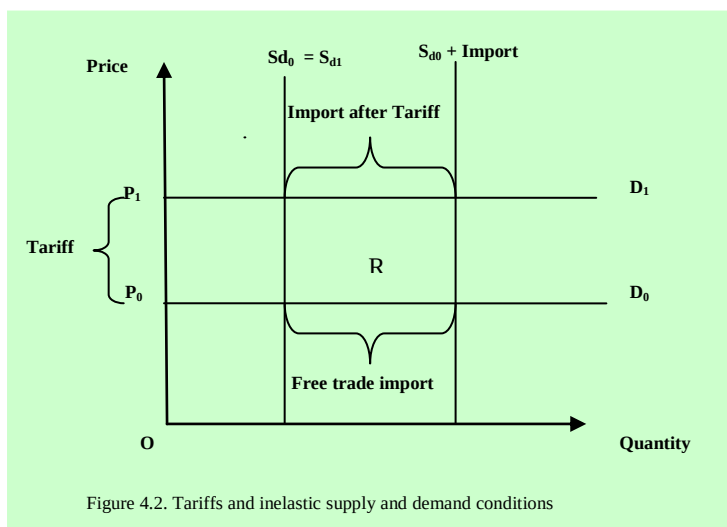
- 1) they both have the same objectives like protecting domestic industries, correcting BOP, and expand domestic employment and economic activities
- 2) A tariff of a certain height cuts imports to a certain quantity. It has, therefore a quota equivalent effect. At the same time a quota would limit imports to a certain quantity and therefore, raises the import price. A quota has, thus, a tariff equivalent effect.

- 3) Since both quota and tariffs raise the import price and reduce the import quantity they produce similar effects on consumption, production, trade balance, TOT, national income redistribution, factor movements, economic growth and economic welfare.

b. differences

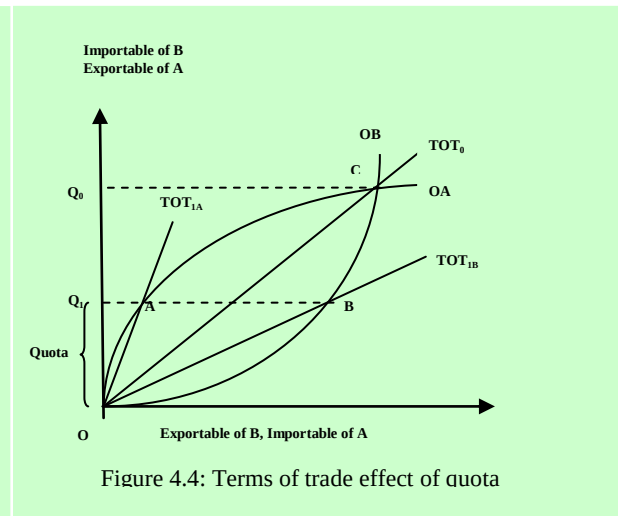
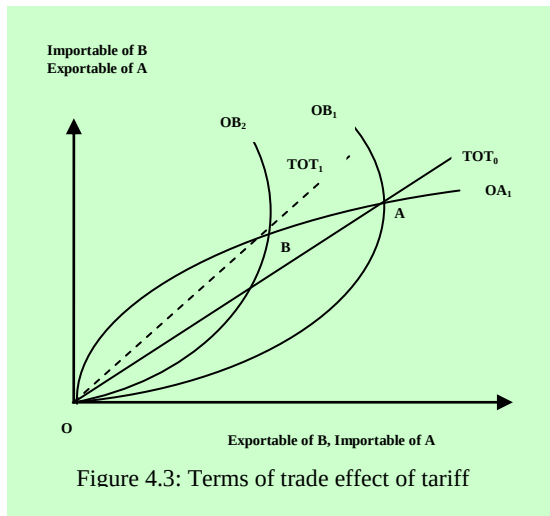
1. Tariffs bring revenue to the government whereas quotas do not. Under tariff same of consumers' loss goes to government in the form of revenue. But under quota it is ambiguous. It could go to government if it charges a fee for selling import licenses and if not, some of consumers loss will be distribute to importers and the imports welfare will increase further more. Tariff revenues can be used for investment on social services, but the quota profits going to importers may not contribute to net social welfare.
2. Distribution of import licenses (associated with quota) may give rise to corruption and bribery on the part of government. Import tariffs do not create such problems or even if they do the degree is small.

Quotas could be more effective than tariffs particularly when the domestic demand and supply curves for the imported good are inelastic. As figure 4.2 indicates the demand and supply curves are perfectly inelastic. A rise in the price of the commodity from P_0 (free trade price) to P_1 (price after tariff) does not bring any change in the quantity of import. The import quantity before and after tariff are equal. Tariff may of course results in substantial government revenue equal to



the area of rectangle R, but it has failed to bring protection effect. Import quotas, are therefore, more effective in protecting domestic industries that produce products with inelastic demand and supply behavior.

3. **The TOT effects of tariffs are determined or predictable, but those resulting from quotas are unpredictable.**



4. There are several reasons why domestic producers, importers, and even governments may prefer quota to tariffs (refer section 4.2.2).

4.2.2.2 Subsidies

National governments sometimes grant subsidies to domestic producers to help improve their trade position. Such devices are an indirect form of protection provided to home businesses, whether they may be import-competing producers or exporters. By providing domestic firms a cost advantage, a subsidy allows them to market their products at prices lower than their actual cost or profit considerations warrant. Governments wanting to see certain domestic industries expand may provide subsidies to encourage their development. Government subsidies assume a variety of forms. In the simplest method, a government makes an outright cash disbursement to a domestic exporter after the sale has been completed. The payment may be according to the discrepancy between the exporter's actual costs and the price received or on the basis of a fixed amount for each unit of a product sold. The overall result is to permit the producer a cost advantage that would not otherwise exist. Such direct export subsidies when applied to manufactured goods have been prohibited by the

General Agreement on Tariffs and Trade.

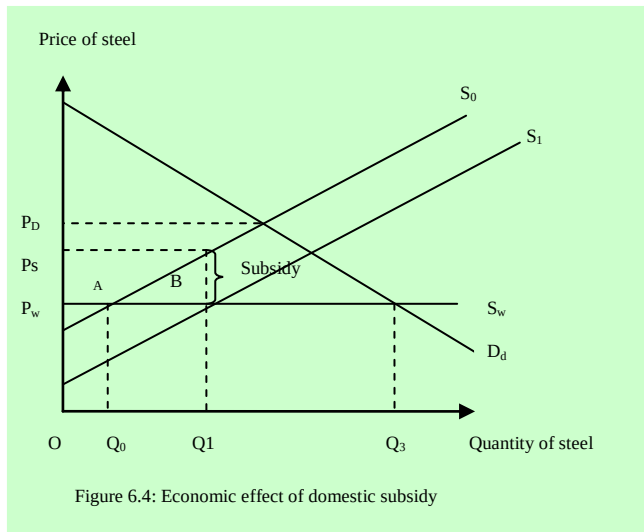
Governments sometimes use indirect subsidies to achieve the same general result. For example, governments may give their exporters special privileges, including tax concessions, insurance arrangements, and loans at below market interest rates. Governments may also sell surplus materials (such as ships) to domestic exporters at favorable prices. Governments may purchase a firm's product at a relatively high price and then dump it in foreign markets at lower prices. This has traditionally been the technique used by the U.S. government in conjunction with its farm-support programs. As with direct cash disbursements to domestic producers, indirect subsidies are intended to encourage the expansion of a nation's exports by permitting them to be sold abroad at lower prices.

For purposes of our discussion, two types of subsidy can be distinguished: a domestic subsidy, which is sometimes granted to producers of import-competing goods, and an export subsidy, which is made to producers of goods that are to be sold overseas. In both cases, the recipient producer views the subsidy as tantamount to a negative tax: the government adds an amount to the price the purchaser pays rather than subtracting from it. The net price actually received by the producer equals the price paid by the purchaser plus the subsidy. The subsidized producer is thus able to supply a greater quantity at each consumer's price.

i. Domestic Subsidy

Figure 6.6 illustrates the trade and welfare effects of a subsidy granted to import-competing producers. Assume that the initial domestic supply and demand for steel are depicted by curves SS_0 and DD_0 so that the market equilibrium price is P_0 per ton.

Assume also that, because the home country is a small buyer of steel, changes in its purchases do not affect the world price of P_w per ton. Given a free-trade price of P_w per ton, the home country consumes Q_3 tons of steel, produces Q_0 tons, and imports $Q_3 - Q_0$ tons. To partially insulate domestic producers from foreign competition, suppose the home country's government grants them a production subsidy of $P_s - P_w$ per ton of steel. The cost advantage made possible by the subsidy results in a shift in the domestic supply curve from SS_0 to SS_1 . Domestic production expands from Q_0 to Q_1 tons, and imports fall from $Q_3 - Q_0$ to $Q_3 - Q_1$ tons. These changes represent the subsidy's trade effect.



The subsidy also affects the national welfare of the home country. According to Figure 6.6, the subsidy permits home country output to rise from Q_0 to Q_1 tons. Note that, at this output, the net price to the steelmaker is P_s —the sum of the price paid by the consumer P_w plus the subsidy $P_s - P_w$. To the home government, the total cost of protecting its steelmakers equals the amount of the subsidy $(P_s - P_w)$ times the amount of output to which it is applied OQ_1 .

Where does this subsidy revenue go? Part of it is redistributed to the more efficient domestic producers in the form of producer surplus. This amount is denoted by area a in the figure. There is also a protective effect, whereby more costly domestic output is allowed to be sold in the market as a result of the subsidy. This is denoted by area b in the figure. To the domestic economy as a whole, the protective effect represents a deadweight loss of welfare.

Attempting to encourage production by its Import-competing producers, a government might levy tariffs or quotas on imports. But tariffs and quotas involve larger sacrifices in national welfare than would occur under an equivalent subsidy. Unlike subsidies, tariffs and quotas distort choices for consumers (resulting in a decrease in the domestic demand for imports), in addition to permitting less efficient home production to occur. The result is the familiar consumption effect of protection, whereby a deadweight loss of consumer surplus is borne by the home nation. This welfare loss is absent in the subsidy case. A subsidy tends to yield the same result for domestic producers as does an equivalent tariff or quota, but at a *lower* cost in terms of national welfare.

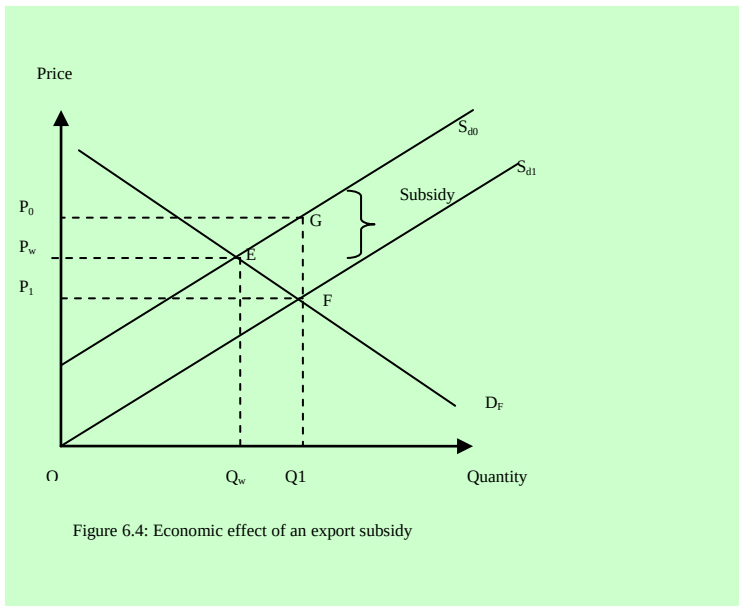
Subsidies are not free goods, however, for they must be financed by someone. The direct cost of the subsidy is a burden that must be financed out of tax revenues paid by the public. Moreover, when a subsidy is given to an industry, it is often in return for accepting government conditions on key matters (such as employee compensation levels). The superiority of a subsidy over other types of commercial policies may thus be less than the preceding analysis suggests.

ii. Export Subsidy

Besides attempting to protect import-competing industries, many national governments grant subsidies, including special tax exemptions and the provision of capital at favored rates, to increase the volume of exports. By providing a cost advantage to domestic producers, such subsidies are intended to encourage a nation's exports by reducing the price paid by foreigners. Foreign consumers are favored over domestic consumers to the extent that the foreign price of a subsidized export is less than the product's domestic price.

The granting of an export subsidy yields two direct effects for the home economy: a *terms-of-trade effect* and an *export revenue effect*. Because subsidies tend to reduce the foreign price of home-nation exports, the home nation's terms trade worsened. But lower foreign prices generally stimulate export volume. Should the foreign demand for exports be relatively elastic, so that a given percentage drop in foreign price is more than offset by the rise in export volume, the home nation's export revenues would increase.

Figure 6.7 illustrates the case of an export subsidy applied to a commodity in trade between home country and a foreign country. Under free trade, market equilibrium exists at 'point *E*', where the home country exports Q_0 units of the commodity to the foreign country at a price P_w . Suppose the exporting country government, to encourage export sales, grants to its exporters a subsidy of $P_s P_1$ per commodity. The home country's supply curve will shift from SS_0 to SS_1 and market equilibrium moves to point *F*. The terms of trade thus turn against exporting country because its export price falls from P_w to P_1 per commodity exported.



Whether the exporter country's export revenue rises depends on how foreign country buyers respond to the price decrease. If the percentage increase in the number of the commodity sold to the foreign country's buyers more than offsets the percentage decrease in price, home country's export revenue will rise. This is the case in Figure 6.7, which shows the exporter country's export revenue rising from OP_wEQ_1 to OP_1FQ_1 units of currency as the result of the decline in the price of its export good.

Although export subsidies may benefit industries and workers in a subsidized industry by increasing sales and employment, the benefits may be offset by certain costs that fall on the society as a whole. Consumers in the exporting nation suffer as the international terms of trade moves against them. This is because, given a fall in export prices, a greater number of exports must be exchanged for a given dollar amount in imports. Domestic consumers also find they must pay higher prices than foreigners for the goods they help subsidize. Furthermore, to the extent that taxes are required to finance the export subsidy, domestic consumers find themselves poorer.

CHAPTER SUMMARY

- *Free trade is a policy of no restriction on the movement of goods between countries whereas a policy used to encourage domestic industries by the use of barriers like imposition of high custom duties on foreign products is termed as a protection policy.*
- *Advocates of free trade, mostly the classical economists, argue for free trade based on the dynamic and static gains that countries can get from international trade.*

- There are a number of arguments in favor of protectionism. The arguments are based on the importance of protection for infant industries, for diversification of industries, for increasing domestic employment, for conserving natural resources, for developing key industries, for defense as source of government revenue, to check balance of payment problem, to be self sufficient, to avoid unfair competition between foreign and domestic producers, and to avoid economic instability due to external shocks.
- Barriers to trade are grouped as natural and man made barriers. The man made (artificial) barriers to trade are further classified into tariffs and non-tariff barriers.
- Tariffs can be classified as specific, ad valorem, compound, discriminatory, non-discriminatory, revenue, protective, retaliatory, and countervailing tariffs.
- The major effects of import tariff on the tariff imposing country are: increase in price of the product in the domestic market, decrease in consumption level of domestic consumers, increase in domestic production (import substitution effect), brings revenue to the government, reduction in import payments (BOP effect), decrease in consumers welfare and increase in producers welfare (that leads to a net social welfare loss), favorable terms of trade, and increase in employment and national income.
- Some of the reasons why nations often use non-tariff barriers like quota and subsidies and others than tariffs as a trade policy include political reasons (to get vote from the public during election), certainty of restrictive impact of non-tariff barriers (especially quota) compared to tariffs, and the fact that non-tariff barriers are not covered by rules of the WTO (GATT).
- Both quota and tariff raise the import price and reduce import quantity. As a result, they produce similar effects on consumption, production, trade balance, terms of trade, national income redistribution, economic growth, and economic welfare.
- Tariff and quota are different in terms: revenue (tariff brings revenue to government but quota not), degree of risk of corruption (import quota licensing is more susceptible for risk of corruption than tariff), quota could be more effective in protecting domestic producers that produce a product whose supply is high or perfectly price inelastic, and the terms of trade effect of tariff is predictable but that of quota is unpredictable.
- Governments may provide subsidies to domestic producers that produce import competing products as domestic subsidy and to domestic producers that produce for export as exporter subsidies.

REVIEW QUESTIONS

1. Discuss the arguments against free trade.
2. Discuss the arguments against protectionism
3. Explain the effects of import tariff on the tariff imposing country with the help of graph.
4. Unilateral quotas can be global or allocated. Explain the basic features of global and allocated quotas.
5. Write the similarities and differences between import tariff and import quota.
6. Explain the economic effects of domestic subsidy given to import competing producers and export subsidy using graphs.
7. A sudden shift from import tariffs to free trade may induce short-term unemployment in Import-competing industries. **Why & how?**

CHAPTER SIX**6. INTERNATIONAL ECONOMIC COOPERATIONS AND ORGANIZATIONS****CHAPTER OBJECTIVES**

The objective of this chapter is to acquaint students with the merits and demerits of international economic cooperation and the functions of international organizations like the world Bank, International Monetary Fund and World Trade Organization.. After learning the contents in this chapter, students will be able to:

- *Understand the basic reason or advantages of international and regional economic cooperation between countries.*
- *Identify the different forms of international economic cooperation and differentiate them according to their scope.*
- *Be familiarized with the history and basic functions of international organizations like the World Bank, International Monetary Fund(IMF), and the World Trade Organization (WTO).*

6.1 Why International Economic Co-operations?

International economic relations imply a mutual economic influence between countries. Exports of goods, international capital flows or changes in one country's economic policy will always influence another country to a greater or lesser degree.

International economic cooperation is prompted mainly by this mutual influence which countries have via their economic policy and subsequently through their international transactions. The 'spill-over' of one country's economic policy on the economies of other countries is usually an unintentional side effect. In welfare theory it is called an *external effect*. The consequence of such an external effect is that, from the point of view of total world welfare, the country pursuing this economic policy is not applying the correct dose, because it is concerned only with the costs and benefits which it will itself derive from that policy. These determine the exact form and extent of the policy devised. The ensuing costs and benefits in another country are ignored. If the policy makers were to take these into

account as well, then they would have undoubtedly devised a different optimum policy as being the optimum policy.

The aim of international economic cooperation is that national economic policy decisions should also take account of their effects on other countries' economies. This seldom if ever seems to happen spontaneously. National governments obviously focus mainly on their own country. That is what they are for, because that is the location of the electorate who will ultimately decide whether to re-elect the policy makers. .

9.2 Forms of international economic cooperation

The forms of international economic cooperation can be categorized into four based on the degree of closeness of countries making the economic cooperation. These are cooperation for exchange of information, international economic policy coordination, international policy harmonization, and economic unification or integration.

a) Exchange of information

There has been some degree of international economic cooperation for hundreds of years. Often it was in a very moderate form, namely the exchange of information on the state of the countries' own economies and on their plans for future national economic policy. It was not until this century that international economic cooperation became more or less permanent. At the end of the Second World War it actually gained momentum.

Since the Second World War the *exchange of information* on national economic policy has been incorporated in the work of the Organization for Economic Cooperation and Development (OECD), which has its secretariat in Paris. Government representatives regularly meet there to exchange ideas on the economic situation in their country and forthcoming economic policy. With the information thus obtained, a country's government is in a better position, when preparing its policy, to take account of the expected policy in other countries and the influence which that is likely to have on the economy of the home country. For example, if it is apparent that Germany is preparing for an expansionary policy, the governments of countries such as Belgium and the Netherlands can react by moderating any plans which they have for expansion. They can expect that via the increased German demand for imports, stronger growth in Germany will have an expansionary influence on their economies, which traditionally send a substantial percentage of their exports to

b) International policy coordination

A closer form of international cooperation is *policy coordination*. All countries involved in the coordination have their own preferences as regards the future values of their economic policy objectives. They are aware of the mutual influence exerted by national economic policy. The ultimate aim of policy coordination is that individual countries should adjust their policy instruments in the best interests of all the economic objectives of the countries concerned. In theoretical form: the econometric model containing all the countries concerned includes all national economic objectives and economic policy instruments; there is also an objective function comprising all countries and containing all national objectives, each in relation to the target value for that objective, with a specific weighting which expresses the importance of the objective concerned. The aim is to maximize the objective function under the condition of the econometric model. This process yields the optimum value of each of the national policy instruments, in the best interests of the group of countries concerned - although it cannot be precluded that the result of this optimization is that some countries' welfare declines.

Since the mid 1970s, international policy coordination has taken place mainly in the G-7. This is the group of seven leading industrial countries: the US, Japan, Germany, United Kingdom (UK), France, Italy and Canada. Since 1976 these G-7 members have held annual meetings of the heads of state or government leaders of the participating countries. The agreements made at these meetings are general in character, aiming, for instance, to reduce the American budget deficit, cut-taxes in Japan and produce a more expansive government policy in Germany. There is no retrospective monitoring or evaluation. The agreed policy often fails to be implemented because, according to the country concerned, new developments during the year necessitated interim adjustments of their policy intentions.

Various other groups of countries have developed in addition to the G- 7. These also aim at coordination, but in regard to a specific aspect of economic policy. Thus, the G-10 has been in existence since 1962. It consists of the G- 7 countries plus the Netherlands, Belgium and Sweden, with the addition of Switzerland in 1964 this group became involved in the IMF's liquidity position on the initiative of the US; Since 1972 there has also been a G-24

comprising a large group of developing countries. The number of members rapidly grew to far more than 24. This group discusses international monetary issues and tries to arrive at common positions before the annual meeting of the IMF, so that the developing countries can speak with one voice and thus have more say at that meeting. There have been periods in which a G-5 looked likely to develop as a permanent consultation agency. This group consisted of the G-7 without Italy and Canada. However, the G-5 now seems to have had its day. Instead there is a tendency to establish a G-3 consisting of the US, Japan and Germany, or a variant in which Germany is replaced by the European Union (EU). The G-7 coordination meetings could then become a matter for the G-3.

c) International policy harmonization

Another form of international economic cooperation is international *policy harmonization*. We usually talk of harmonization if the object of cooperation is to achieve a degree of convergence in policy intervention in the various countries. Thus, harmonization does not so much concern macro-economic policy but rather policy intervention which influences the international competitive position of national enterprise, e.g. the setting of rates of tax - particularly indirect taxes such as VAT rates - and government regulations aimed at protecting the safety and health of consumers. This has been attempted in the EC, in particular, in recent years. As regards VAT, so much progress has been made that all countries impose their indirect taxation on the basis of VAT and a minimum level has been set for both the low and high rates of VAT. This system combats price advantages which are not based on low production costs but on something else than comparative advantage such as favorable rates of tax.

d) Unification/integration of economic policy

The most radical form of international economic cooperation is total *integration* or *unification* of economic policy in the countries concerned. Obviously, such an economic policy is *supranational* in practice. This means the existence of economic control at a higher than national level, mapping out and implementing economic policy for the countries involved in the unification. Another form of supranational decision-making is majority decisions by participating countries. This form of cooperation combines well with democratic decision-making, though this requires a supranational parliament. The plans for an Economic and Monetary Union (EMU) in the EU clearly contain supranational elements

The above four forms of international economic cooperation were put in ascending order of closeness, these are: exchange of information as the least radical form, coordination, harmonization and unification as undoubtedly the closest form of international economic cooperation. For each of these a further distinction can be made according to the scope of the cooperation, which ranges from strictly sectoral (e.g. international cooperation on trade policy for the textile sector only, in the form of the Multi-Fibre Arrangement) to fully integrated economic policy. There is also a geographical differentiation in scope: from small-scale, regional cooperation (e.g. by Belgium, the Netherlands and Luxembourg in the Benelux) to truly global cooperation (as in the IMF).

Advantages of international economic cooperation

As stated in section 9.1, the advantage of international economic cooperation is undeniably that the effects on other countries' economies are taken into account in determining national policy.

Disadvantages of international economic cooperation

However, in economics there are no advantages without disadvantages; it remains a science based on weighing up costs and benefits, and international economic cooperation is no exception to that. The disadvantages of international cooperation can be divided into: the sacrifice of national autonomy in policy-making; the complexity of implementation; the risk of cheating; and free riders. These four disadvantages will now be explained in more detail.

a) Loss of national autonomy in policy making

Loss of national autonomy in the implementation of economic policy is often in itself a psychological disadvantage for a national government. Independence and autonomy are usually regarded as positive attributes. But optimum international economic cooperation ought to lead to maximum economic welfare for the whole co-operating region. However, one cannot rule out the possibility that part of the region, e.g. one of the co-operating countries, will nevertheless suffer a loss of welfare or gain hardly anything at all. Much depends on the weightings given to that country's economic objectives in the overall economic welfare function. Those weightings will be small if the country in question is of minor importance in the overall co-operative framework. In economic terms, it is the largest countries which have the power. They have very little interest in cooperation since their economy is relatively little influenced by other countries. For small countries the exact opposite applies. As a result, the large countries will be able to impose the most

demands, and they will do so.

b) Complexity of implementation

International economic cooperation in the form of policy coordination is *difficult to implement*. It is in fact even more difficult to implement than national economic policy. In formal terms, ascertaining the outcome of optimum international policy coordination demands knowledge of a number of elements, which are particularly difficult to quantify. One is the combined welfare function for the countries taking part in the coordination. Another is the composition of national models considered to give a good quantitative description of the participating economies. At national level this information is already hard to come by, and people continue to disagree on the correct form. Where several countries are concerned, the level of disagreement can clearly increase significantly. One complication here is that simulation results of econometric models for policy coordination indicate that use of a model which, on closer examination, proves to be incorrect or unsatisfactory as a reflection of the economy involved, can easily more than negate the positive result of coordination.

c) The risk of cheating or behaving as free riders

The last issue in connection with policy coordination concerns the fact that (potential) participants withdraw from the agreements or from the cooperation as a whole. In this case, the cooperation is undermined by participants *cheating* or behaving as *free riders*. Once the form in which the coordinated economic policy is to be implemented has been decided by mutual consultation, it can be to the advantage of a participating country to fail to carry out its own part. The other countries' policy will in itself make a positive contribution to the desired economic development of the country in question. One reason for trying to avoid participating in the implementation of the policy may be that the country's own intended contribution to the coordinated policy entails costs for that country: One example is the implementation of an expansive fiscal policy which has to be financed on the capital market by issuing government bonds. Failure to meet obligations, or cheating, cannot be avoided altogether; at best the risk can be reduced. In this connection, retrospective legal action serves no practical purpose, because there are no legal rules for this type of international agreement, although failure to comply with agreements will be less attractive if such policy coordination is not an isolated instance but continues from year to year. Cheating in anyone

year then entails costs for the country concerned, in that it is bound to be excluded from the policy coordination in subsequent years - and hence also from the important phase of devising the common policy! The risk of cheating can be further reduced if the policy for countries taking part in the cooperation is laid down in an agreement and if an organization is set up to supervise compliance with the rules.

The free rider issue concerns the fact that a country can withdraw not only from implementing the coordinated policy but also from devising it. In short, the country remains totally outside the international economic cooperation. Such an attitude combined with the expectation that the country will actually benefit from the common policy of the remaining group of countries - in the form of an improvement in economic development - makes the country into a free rider. Nothing can be done about this problem apart from bringing political pressure to bear on the free rider. A single free rider will not block international policy coordination. However, if a number of mainly larger countries thus stay out of range, this can mean the end of the attempt at coordination.

9.3 Regional integration

Ever since the creation of the European Economic Community (EEC), in 1958, regional groupings, have despite the reservations of many economists, become ever more popular. In fact, one has the impression that the world is turning into a mass of regional groupings. These may take any of the following forms, a co-operation agreement, a free trade agreement, a customs union, a common market and/or an economic and monetary union, customs union, a common market and/or an economic and monetary union (EMU).

It is generally accepted that countries at a particular level of economic and trade development will choose one particular form of regional integration rather than another. However, there may be political considerations of a sometimes longer-term nature which may persuade some countries to pay a short- or medium-term economic and financial price in exchange for longer-term benefits. Thus, for example, last century, when the German-speaking states united to create Germany, Prussia allowed the smaller states to enjoy benefits knowing that in the long run Prussia would dominate Germany. Similarly, at the present time, in the European Union, Germany, albeit grudgingly, makes very large financial contributions to the Community budget because she knows that she enjoys a captive market

inside the union.

Despite the existence of such examples, economists, ever since the publication of Viner's pathbreaking book in 1950, "The Customs Union Issue", have been puzzled as to why countries should, for example, join a customs union rather than orgasms non-preferential multilateral trade agreements. At the two extremes, EI-Agaa, writing in 19892, lists clear reasons why such regional groupings may appear to be advantageous to countries. These include the following:

- Enhanced efficiency as the result of specialization
- Increased production levels as a result of the exploitation of economies of scale
- Improved international bargaining positions as a result of increased size (and a long run improvement in the terms of trade)
- Enforced efficiency gains as a result of increased competition, and
- Increased rate of growth as a result of technological advances.

9.4 The World Bank, International Monetary Fund and World Trade Organization

9.4.1 The World Bank

The World Bank is like a cooperative, where its 184 member countries are shareholders. The shareholders are represented by a Board of Governors, who are the ultimate policy makers at the World Bank. Generally, the governors are member countries' ministers of finance or ministers of development.

Conceived during World War II at Bretton Woods, New Hampshire, the World Bank initially helped rebuild Europe after the war. Its first loan of \$250 million was to France in 1947 for post-war reconstruction. Reconstruction has remained an important focus of the Bank's work, given the natural disasters, humanitarian emergencies, and post conflict rehabilitation needs that affect developing and transition economies.

Nowadays the Bank, however, has sharpened its focus on poverty reduction as the overarching goal of all its work. It once had a homogeneous staff of engineers and financial analysts, based solely in Washington, D.C. Today, it has a multidisciplinary and diverse staff including economists, public policy experts, sectoral experts, and social scientists. 40 percent of staff is now based in country offices.

The Bank itself is bigger, broader, and far more complex. It has become a Group, encompassing five closely associated development institutions:

- the International Bank for Reconstruction and Development (IBRD),
- the International Development Association (IDA),
- the International Finance Corporation (IFC),
- the Multilateral Investment Guarantee Agency (MIGA), and
- the International Centre for Settlement of Investment Disputes (ICSID).

The World Bank's two closely affiliated entities—the International Bank for Reconstruction and Development (IBRD) and the International Development Association (IDA)—provide low or no interest loans and grants to countries that have unfavorable or no interest loans and grants to countries that have unfavorable or no access to international credit markets

Fund Generation

IBRD lending to developing countries is primarily financed by selling bonds in the world's financial markets. While IBRD earns a small margin on this lending, the greater proportion of its income comes from lending out its own capital. This capital consists of reserves built up over the years and money paid in from the bank's 184 member country shareholders. IBRD's income also pays for World Bank operating expenses and has contributed to IDA and debt relief.

IDA, the world's largest source of interest-free loans and grant assistance to the poorest countries, is replenished every three years by 40 donor countries. Additional funds are regenerated through repayments of loan principal on 35-to-40-year, no-interest loans, which are then available for re-lending.

Loans

Through the IBRD and IDA, the bank offers two basic types of loans and credits.

- Investment loans: Investment loans are made to countries for goods, works and services in support of economic and social development projects in a broad range of economic and social sectors.

- Development policy loans: Development policy loans (formerly known as adjustment loans) provide quick-disbursing financing to support countries' policy and institutional reforms

Grants

Grants are designed to facilitate development projects by encouraging innovation, co-operation between organizations and local stakeholders' participation in projects. For example in recent years, IDA grants—which are either funded directly or managed through partnerships—have been used to:

- Relieve the debt burden of heavily indebted poor countries
- Improve sanitation and water supplies
- Support vaccination and immunization programs to reduce the incidence of communicable diseases like malaria
- Support vaccination and immunization programs to reduce the incidence of communicable diseases like malaria
- Combat the HIV/AIDS pandemic
- Support civil society organizations
- Create initiatives to cut the emission of greenhouse gasses

Analytic and Advisory Services

While the bank is best known as a financier, another of its roles is to provide analysis, advice and information to member countries so they can deliver the lasting economic and social improvements their people need. The bank does this in several ways: through economic research on broad issues such as the environment, poverty, trade and globalization and through country-specific economic and sector work, where the bank evaluates a country's economic prospects by examining its banking systems and financial markets, as well as trade, infrastructure, poverty and social safety net issues, for example.

Capacity building

Another core bank function is to increase the capabilities of its staff, partners and the people in developing countries—to help them acquire the knowledge and skills they need to provide

technical assistance, improve government performance and delivery of services, promote economic growth and sustain poverty reduction programs. Linkages to knowledge-sharing networks such as these have been set up by the bank to address the vast needs for information and dialogue about development.

9.4.2 International Monetary Fund (IMF)

The IMF was born at the end of World War II, out of the Bretton Woods Conference in 1945. The Fund was created out of a need to prevent economic crises like the Great Depression. The main purposes for which the I.M.F. was set up were to provide exchange stability, temporary assistance to countries falling short of foreign exchange and international sponsoring of measures for curing fundamental causes of disequilibrium in balance of payments. The I.M.F. is a pool of central -bank reserves and national currencies which are available to its members under certain conditions. It can be regarded as an extension of the central bank reserves of the member countries.

The IMF works to the creation of financial markets worldwide and to the growth of developing countries. With its sister organization, the World Bank, the IMF is the largest public lender of funds in the world. It is a specialized agency of the United Nations and is run by its 184 member states. Membership is open to any country that conducts foreign policy and accepts the statutes of the Fund.

Objectives of IMF

According to Article I of the Fund, the main purposes of the Fund are:

1. To promote international monetary cooperation through a permanent institution.
2. To facilitate the expansion and balanced growth of international trade, and to contribute thereby to the promotion and maintenance of high, levels of employment of the member-countries.
3. To promote exchange stability, to maintain orderly exchange arrangements among members, and to avoid competitive exchange depreciation.
4. To assist in the establishment of a multilateral system of payments in respect of current' transaction between members and in the elimination of foreign exchange restrictions. .
5. To give confidence to members by making the Fund's resources available to them under adequate safeguards, thus providing them with opportunity to correct

maladjustments in their balance of payments without resorting to measures destructive of national or international prosperity (*e.g.*, deflationary policies).

6. In accordance with the above, to shorten the duration and lessen the degree of disequilibrium in the international balance of payments of members.

The IMF is responsible for the creation and maintenance of the international monetary system, the system by which international payments among countries take place. It thus strives to provide a systematic mechanism for foreign exchange transactions in order to foster investment and promote balanced global economic trade.

How Does It Work?

The IMF gets its money from quota subscriptions paid by member states. The size of each quota is determined by how much each government can pay according to the size of its economy. The quota in turn determines the weight each country has within the IMF - and hence its voting rights - as well as how much financing it can receive from the IMF.

Twenty-five percent of each country's quota is paid in the form of special drawing rights (SDRs), which are a claim on the freely usable currencies of IMF members. Before SDRs, the Bretton Woods system had been based on a fixed exchange rate, and it was feared that there would not be enough reserves to finance global economic growth. Therefore, in 1968, the IMF created the SDRs, which is a kind of international reserve asset. It was created to supplement the international reserves of the time, which were gold and the U.S. dollar. The SDR is not a currency; it is a unit of account by which member states can exchange with one another in order to settle international accounts. The SDR can also be used in exchange for other freely-traded currencies of IMF members. A country may do this when it has a deficit and needs more foreign currency to pay its international obligations.

The SDR's value lies in the fact that member states commit to honor their obligations to use and accept SDRs. Each member country is assigned a certain amount of SDRs based on how much the country contributes to the Fund (which is based on the size of the country's economy). However, the need for SDRs lessened when major economies dropped the fixed exchange rate and opted for floating rates instead. The IMF does all of its accounting in SDRs, and commercial banks accept SDR denominated accounts. The value of the SDR is

adjusted daily against a basket of currencies, which currently includes the U.S. dollar, the Japanese yen, the euro, and the British pound.

The larger the country, the larger its contribution; thus the U.S. contributes about 18% of total quotas while the Seychelles Islands contribute a modest 0.004%. If called upon by the IMF, a country can pay the rest of its quota in its local currency. The IMF may also borrow funds, if necessary, under two separate agreements with member countries. In total, it has SDR 212 billion (USD 290 billion) in quotas and SDR 34 billion (USD 46 billion) available to borrow.

Functions of the IMF

The IMF performs five major functions:

1. It serves as a short-term credit institution. If any country is in a temporary difficulty in liquidating an adverse balance of payments, the Fund will come to its aid. It does not, however, undertake to supply all the foreign exchange that a country may need. All countries are supposed to have their separate monetary and foreign exchange reserves to meet their normal requirements. The Fund is not intended to supplant them but to provide only a second line of defence in case of emergency. The borrowing country has to pay interest and maintain its quota intact. Should a country borrow unnecessarily, the rate of interest rises as the amount of loan increases. Lower rates are charged if a loan is taken for a short period. If the amount of the loan and its duration are such as to raise the rate of interest to 5 per cent, the Fund can then raise the rate to any level by way of penalty, for this is regarded as an abuse of the privilege of membership.

Thus; it is clear that the credit operations of the Fund are not only conducted on sound business principle but they also ensure that the object of the Fund, viz., to provide short-term loan only, is not defeated.

2. The Fund provides a mechanism for improving short-term balance of payments position. For this purpose, its rules provide for orderly adjustment of, exchange. No member-country can indulge in irresponsible and competitive exchange depreciation thus introducing the law of the jungle in international monetary relations. Whenever a country feels, that its rate of exchange is out of line with its economy, the rate can be altered but only after due deliberation between the authorities and the authorities of the Fund. There is thus provision for the careful determination of the initial rate and its orderly adjustment subsequently. This procedure at once reconciles the claims to internal

stability and full employment on the one hand and to international stability and high level of world trade on the other. Every country must now rely on its own productive efficiency rather than on artificial stimulus of exchange depreciation to hold its own in the world markets.

3. The Fund provides machinery for international consultations. It brings together representatives of the principal countries of the world and affords an excellent opportunity for reconciling their conflicting claims. This constructive approach and the measure of international co-operation have had not only a stabilizing influence on world economy but they have also led to the expansion and balanced development of world trade and world production. The Fund has thus contributed to the promotion and maintenance of high levels of employment and real income and to the development of the productive resources of the member countries. For this purpose, the Fund is engaged in constant study and research relating to the important and urgent economic problems of the world.
4. It provides a reservoir of the currencies of the member-countries and enables members to borrow one another's currency.
5. It promotes orderly adjustment of exchange rates to promote exchange stability.

9.4.3 World Trade Organization (WTO)

The World Trade Organization (WTO) is an international, multinational organization, which sets the rules for the global trading system and resolves disputes between its member states; all of whom are signatories to its approximately 30 agreements.

The Bretton Woods Conference of 1944 proposed the creation of an International Trade Organization (ITO) to establish rules and regulations for trade between countries. Members of the UN Conference on Trade and Employment in Havana agreed to the ITO charter in March 1948, but ratification was blocked by the U.S. Senate (WTO, 2004b). Some historians have argued that the failure may have resulted from fears within the American business community that the International Trade Organization could be used to regulate (rather than liberate) big business (Lisa Wilkins, 1997; Helen Milner 1993).

Only one element of the ITO survived: the General Agreement on Tariffs and Trade (GATT). Seven rounds of negotiations occurred under the GATT before the eighth round -

known as the Uruguay Round - which began in 1984 and concluded in 1995 with the establishment of the WTO. The GATT principles and agreements were adopted by the WTO, which was charged with administering and extending them and approximately 30 other agreements and resolving trade disputes between member countries. Unlike the GATT, the WTO has a substantial institutional structure. The WTO aims to increase international trade by promoting lower trade barriers and providing a platform for the negotiation of trade and to their business.

Principles of the trading system

The WTO discussions follow the following fundamental principles of trading.

1. A trading system should be free of discrimination in the sense that one country cannot privilege a particular trading partner above others within the system, nor can it discriminate against foreign products and services.
2. A trading system should tend toward more freedom, that is, toward fewer trade barriers (tariffs and non-tariff barriers).
3. A trading system should be predictable, with foreign companies and governments reassured that trade barriers will not be raised arbitrarily and that markets will remain open.
4. A trading system should tend toward greater competition.
5. A trading system should be more accommodating for less developed countries, giving them more time to adjust, greater flexibility, and more privileges

The WTO has 149 members (76 members at its foundation and a further 73 members joined over the following ten years). The 25 states of the European Union are represented also as the European Communities. Some non-sovereign autonomous entities of member states are included as separate members.

A number of non-members, around 31 countries including Ethiopia, have been observers at the WTO and are currently negotiating their membership. The WTO oversees about 30 different agreements which have the status of international legal texts. Member countries must sign and ratify all WTO agreements on accession. The following are some of the most important agreements of WTO.

1) Agreement on Agriculture (AoA)

The AoA came into effect with the establishment of the WTO at the beginning of 1995. The AoA has three central concepts, or "pillars": domestic support, market access and export subsidies.

Criticism: The AoA is criticized for reducing tariff protections for small farmers – a key source of income for developing countries – while allowing rich countries to continue to pay their farmers massive subsidies which developing countries cannot afford.

2) General Agreement on Trade in Services (GATS)

3) Trade-Related Aspects of Intellectual Property Rights (TRIPs) Agreement

4) Sanitary and Phyto-Sanitary (SPS) Agreement

The Agreement on the Application of Sanitary and Phytosanitary Measures - also known as the SPS Agreement was negotiated during the Uruguay Round of the General Agreement on Tariffs and Trade, and entered into force with the establishment of the WTO at the beginning of 1995.

Under the SPS agreement, the WTO sets constraints on member-states' policies relating to food safety (bacterial contaminants, pesticides, inspection and labelling) as well as animal and plant health (imported pests and diseases).

SPS & Genetically Modified Organisms (GMOs)

In 2003, the USA challenged a number of EU laws restricting the importation of Genetically Modified Organisms (GMOs), arguing they are “unjustifiable” and illegal under SPS agreement. In May 2006, the WTO's dispute resolution panel issued a complex ruling which took issue with some aspect's of the EU's regulation of GMOs, but dismissed many of the claims made by the USA.

Quarantine policies play an important role in ensuring the protection of human, animal and plant health. Yet under the SPS agreement, quarantine barriers can be a 'technical trade barrier' used to keep out foreign competitors.

The SPS agreement gives the WTO the power to override a country's use of the precautionary principle – a principle which allows them to act on the side of caution if there is no scientific certainty about potential threats to human health and the environment. Under SPS rules, the burden of proof is on countries to demonstrate scientifically that something is dangerous before it can be regulated, even though scientists agree that it is impossible to predict all forms of damage posed by insects or pest plants.

5) Agreement on Technical Barriers to Trade (TBT)

The Agreement on Technical Barriers to Trade - also known as the TBT Agreement is an international treaty of the World Trade Organization. It was negotiated during the Uruguay Round of the General Agreement on Tariffs and Trade, and entered into force with the establishment of the WTO at the beginning of 1995.

The object of the TBT Agreement is to "to ensure that technical negotiations and standards, as well as testing and certification procedures, do not create unnecessary obstacles to trade.

CHAPTER SUMMARY

- The aim of international economic cooperation is that national economic policy decisions should also take account of their effects on other countries' economies.
- The different forms of international economic cooperation in ascending order of closeness between the countries are: exchange of information, coordination, harmonization, and unification.
- The disadvantages of international economic cooperation can be grouped into: the sacrifice of national autonomy in policy making; the complexity of implementation; and the risk of cheating and free riders. The reasons why countries for regional integrations or groupings are countries get: enhanced efficiency as the result of specialization; increased production levels as a result of the exploitation of economies of scale; improved international bargaining positions as a result of increased size (and a long run

improvement in the terms of trade); enforced efficiency gains as a result of increased competition; and Increased rate of growth as a result of technological advances.

- The World Bank encompasses five closely associated development institutions. These are: IBRD, IDA, IFC, MIGA and ICSID.
- The World Bank uses the funds it generates to provide investment loans and development policy loans to its member countries through IBRD and IDA. It also provide grants to facilitate development projects by encouraging innovations and cooperation between organizations and local stakeholders' participation in projects
- The IMF is responsible for the creation and maintenance of the international monetary system, the system by which international payments among countries take place. It thus strives to provide a systematic mechanism for foreign exchange transactions in order to foster investment and promote balanced global economic trade.
- The World Trade Organization is an international, multinational, organization, which sets the rules for the global trading system and resolves disputes between its member states; all of whom are signatories to its approximately 30 agreements.

REVIEW QUESTIONS

1. Discuss why countries need international economic cooperation.
2. Explain the different forms of international economic cooperation.
3. List and discuss the disadvantages of international economic cooperation.
4. Write the reasons why countries form regional groupings or integrations.
5. List the five development institutions of the World Bank group.
6. What are the two basic types of loans and credits of the World Bank that are extended to member countries through IBRD and IDA?
7. Give examples of IDA grants.
8. Write the objectives of IMF
9. Discuss the major functions of IMF.
10. Write the five principles of trading system followed by the WTO discussions.
11. List the five fundamental agreements of WTO and write the three central concepts of AoA.

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Course Name: International Economics I**Course Code: Econ 3081.****Course Description**

This Course contains two courses: International Economics I and International Economics II. These courses deal with international trade theories and policy issues and how international trade is financed. The Course aims to provide a highly focused discussion of topics in international economics in two separate subject areas of International trade and international finance in the context of developing countries. Issues covered in the Course are conventional and nonconventional trade theories which include the classical model (Smith and Ricardo), the modern and neoclassical theory of trade (the H-O-S model, the Stolper-Samuelson Theorem, and The Leontief Paradox), The New Trade Theories and the African context (Imperfect competition and scale economies), Economic Integration and Theories of Customs Union, Trade Policies and Developing countries (Import Substitution and Export Promotion, Trade and Development, the World Trade Organization and Developing Countries, etc). Besides, the issues of the Balance of Payments (and different approaches to balance of payments), Exchange rate economics, Open-Economy macroeconomics (IS-LM-BP framework, the Mundel-Fleming Model), the Evolution of International Financial Institutions (the IMF, the World Bank etc...), impact of capital flows and the debt crisis in African context will be covered.

Course Objective**The objective of this Course is to:**

- ✓ Demonstrate understanding of the various effects of trade policy interventions
- ✓ Be able to identify those groups that benefit and lose from a move from free to restricted trade
- ✓ Demonstrate how to apply economic reasoning to global policy issues in a critical manner
- ✓ Demonstrate knowledge of the current issues in economics, an appreciation of how economists address international economic issues and be familiar with current research issues in international economics
- ✓ Understand the economic basis behind current policy debates in international economics
- ✓ Critically address different policy proposals in addressing issues arising in international economics
- ✓ Deliver coherent argument in written work, including coherent arguments for assessing complex policy issues in the global economy

International economics I***Module I***

- ✓ Develop presentation skills in addressing theoretical and empirical issues in international economics
- ✓ Lay a foundation of open Macro International Economics
- ✓ Discuss and analyze nation's international interaction through trade and finance to the rest of the world.
- ✓ Explain approaches and techniques of analyzing trade and finance internationally.

4 hrs	I. INTRODUCTION 1.1. Meaning, Nature, and Scope of International Economics 1.2. The Components of International Economics 1.3. The Importance of International Economics 1.4. International Economics and Economic Theories 1.5. Purposes of Economic Theories and/or Models 1.6. Testing a Hypothesis/Theory/ Model 1.7. The Role of Assumptions 1.8. Summary and Conclusions	
14 hrs	II INTERNATIONAL TRADE THEORIES 2.1. Historical Development of Modern Trade Theory 2.1.1. Mercantilists' View on Trade 2.1.2. Physiocracy: A Link between Mercantilism and Classical School 2.1.3. The Classical /Traditional Trade Theories 2.1.4. The Neoclassical Trade Theories 2.1.5. The New/ Modern Trade Theories 2.1.4.1. The Point of Departure from the Traditional/Classical trade Theories 2.1.4.2. The Emergence and Development of the New Trade Theories 2.2. Evaluation of Trade Theories 2.3. Summary and Conclusions	
10 hrs	III. INTERNATIONAL TRADE POLICY 3.1. The Instruments of Trade Policy 3.2. Arguments in Favor of and Against Trade Protection 3.3. Trade Policy and Economic Welfare 3.4. Optimal Trade Policy Intervention 3.5. The Process of Trade Liberalization 3.6. The Political Economy of Protection 3.7. Intellectual Property and Intellectual Property Rights 3.8. Summary and Conclusions	

8 hrs	IV. INTERNATIONAL TRADE AND ECONOMIC DEVELOPMENT 4.1. Trade Theory and Economic Development 4.2. Beneficial Effects of Trade 4.3. Static versus Dynamic Analysis in Trade 4.4. Alternative Trade Strategies 4.5. Trade Strategies and Economic Performance 4.6. Export Pessimism and the Prebisch – Singer Thesis 4.7. Trade, the Environment, and Natural Resources 4.8. The Process of Trade Liberalization and its Implications 4.9. The Role of the State: Experiences of South – East Asian Countries 4.10. Current Problems Facing the Developing Countries 4.11. Summary and Conclusions	
12 hrs	V. INTERNATIONAL TRADE INSTITUTIONS AND REGIONAL ECONOMIC ARRANGEMENTS 5.1. Historical Background: From the General Agreement on Tariffs and Trade (GATT, 1947) to World Trade Organization (WTO, 1995) 5.2. WTO Opportunities 5.3. WTO Accession Process and its Challenges 5.4. Issues of Globalization 5.5. Regional Economic Cooperation and Integration 5.5.1. The Definition of Basic Concepts 5.5.2. The Rational for Integration Schemes 5.5.3. The Forms of Integration 5.5.4. The Theory of Customs Union (CU) and its Limitations 5.5.4.1. Trade Creation and Trade Diversion 5.5.5. The Status of Regional Economic Integration in Africa 5.5.6. The Lessons of Integration Experience 5.5.7. Regionalism versus Multilateralism 5.5.8. Summary and Conclusions	

Course Delivery Methods

The delivery method shall be student-centered. Students are highly expected to participate in class works at the middle and end of each session and in group discussions inside and outside of the class. Specifically the course will be delivered through the following methods:

- Lecture Method
- In-class problem solving
- Group Work
- Assignment

Assessment Methods

International economics I

Module I

Student evaluation in this Course consist both formative and summative assessments including quizzes, test and final exam. Marks will be allocated according to the following grading schedule.

Assessment method	Weight
Assignments (Individual / Group)	25%
Mid - Semester	25%
Final Exam	50 %
Total	100%

Course Policy

- ☛ Late coming is not allowed and no student is allowed to enter after class has started.
- ☛ Duplication of assignments is strictly forbidden; it entails serious penalty.
- ☛ Assignments are required to be submitted before or on the deadline.
- ☛ Cheating during exam sessions results in a minimum of “F “grade while cheating in quizzes and tests is subjected to a zero mark. All cheating cases will be reported to the department for further considerations.
- ☛ Students should switch off their cell phones while they are in class and must keep their cell phones switched off during all kinds of exam sessions.
- ☛ Students must attend 80% of the class for the course. Failure to attend 80% of the class will not allow the student to sit for the final exam.
- ☛ Missing a quiz without convincing evidences will earn the students a grade of zero marks in that specific quiz